

Research paper

Ensemble deep learning model for protein secondary structure prediction using NLP metrics and explainable AI

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ABSTRACT

Proteins are a major part of all living organisms, consisting of chains made from amino acids, and they perform various biological processes such as DNA replication, metabolism, or forming cell structure. Detecting accurate protein structures is fundamental to understanding their functions. Advances in deep learning (DL), artificial intelligence, and similar technologies have transformed computational biology. AlphaFold and RoseTTAFold are examples of AI models greatly outperforming traditional methods to predict 3D protein structures with very high accuracy. This study builds on the progress, using an ensemble deep learning model for protein secondary structure prediction. When it came to training and validation, the ensemble model outscored the individual models with the best accuracy (94.41%) and the lowest validation loss (0.1585). Since it maintains the structural integrity of protein sequences, the ROUGE-L score, which is widely used in natural language processing (NLP), works well for protein sequence evaluation. This integrates NLP methods with bioinformatics. Model predictions were made more interpretable by utilizing explainable AI approaches like integrated gradients and LIME, which shed light on the characteristics that affect protein shape. By using this method, biases are lessened, and a greater comprehension of the biological connections between protein shapes and sequences is provided.

1. Introduction

A protein is an enzyme formed from one or more longitudinal chains of amino acid residues called polypeptides. They contribute to biological processes, like catalyzing metabolic reactions, replicating DNA, responding to stimuli, providing structure for cells, and organisms sexually reproducing (meiosis) transporting molecules between locations. Fig. 1 shows the four levels of protein structure [1]. The amino acid sequence in its polypeptide chain is referred to as the primary structure of a protein. Protein biosynthesis makes new peptide bonds between the carboxyl and amino groups of two adjacent monomers in every case. The secondary structure of a protein refers to the local spatial conformation of the continuous amino acid sequence chain (residue) excluding its side chains and is stabilized by hydrogen bond interactions within this contiguous segment. The tertiary structure represents the complete three-dimensional arrangement of the polypeptide chain in the space. Local destabilization is countered by (outside polar) hydrophilic hydrogen bonding and perhaps ionic bond interactions, as well as (internal to the molecule) orienting hydrophobic interactions between non-polar amino acid side chains. The quaternary structure of a protein is the

fourth classification level of protein. The quaternary structure of protein is composed of two or more smaller chains. They participate in many cellular processes as structural components, signaling molecules, and catalysts of metabolic pathways. Proteins act as structural elements, signaling molecules, and catalysts for metabolic events, thus placing them as a crucial need for life to exist. Proteomics refers to the study of proteins and their interactions with other molecules in a cell or system. Recent advances in technology have allowed for the proteome to be discovered [2].

The determination of 3-D protein structure is the basic need for much biological research [3]. Currently, the UniProtKB/TrEMBL database contains over 227 million protein sequences; however, around 50,000 of these sequences have been solved experimentally [4]. There are many traditional methods to determine the protein structures. Some include X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and cryo-electron microscopy. Though powerful, these methods aren't always the feasible and suitable method for the growing demands of protein structure predictions. X-ray crystallography takes months to even years to predict a protein structure. The process of producing, purifying, and crystallizing the protein is often slow and challenging. The chal-

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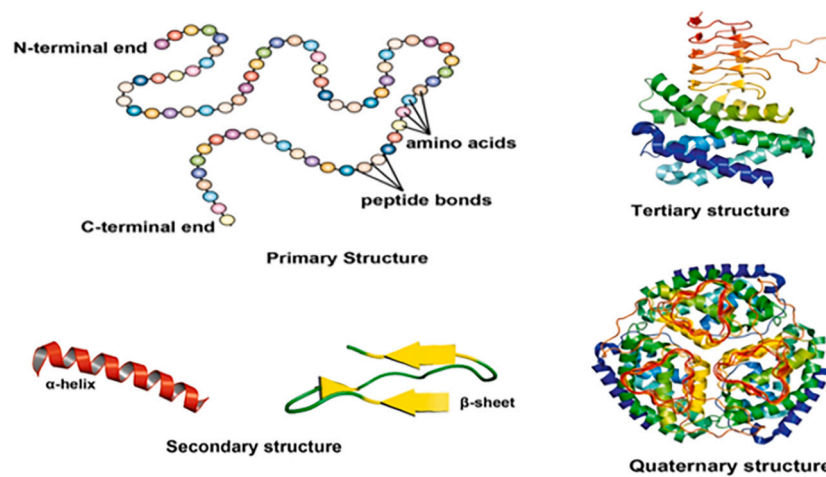


Fig. 1. Four Levels of Protein Structure.

Challenges don't stop here. Crystallization of proteins is not always easy and doable. Large complexes or membrane proteins are difficult or impossible to crystallize [5]. NMR (Nuclear Magnetic Resonance) is a method that allows scientists to observe proteins in a state that's much closer to how they exist in living organisms. This method relies heavily on a collection of experimental structural restraints. This includes NOE-derived local distance restraints (≤ 6 Å), hydrogen bonds, and dihedral angle restraints. NMR structure determination is also limited to small (< 30 kDa) proteins [6,7]. The larger the protein (greater than 50 kDa), the slower it will tumble in solution. This decrease in the rate of tumbling slows down NMR relaxation times, especially T2 (transversal), which results in a faster loss of signals. This decreased sensitivity makes NMR less practical for studying large proteins as the signals become difficult to detect and quantify, thereby complicating the detection of local conformation and distances between regions in the protein [8,9]. Going for traditional methods would need much capital spending on equipment, reagents, and facilities. Furthermore, these experiments are quite challenging to perform and also require a good deal of expertise for proper execution.

For decades, computational biologists have been working on algorithms and computational approaches that can predict protein structures with high accuracy. One of the computational methods is "Homology Modelling". This method uses the fact that evolutionary-related proteins share a similar structure [10]. This modeling is the only method of choice to generate a reliable 3D model of a protein from its amino acid sequence, as notably shown in several meetings of the bi-annual critical assessment of techniques for protein structure prediction (CASP) [11]. The quality of the homology model is directly linked to the identity between the template and target sequences. The models built over 50% sequence similarities are accurate enough for drug discovery applications; those between 25 and 50% identities can help the design of mutagenesis experiments; and those between 10 and 25% are tentative at superlatives [12,13]. Homology modeling depends heavily on the availability of suitable template structures. If no structures are available, the predictions can be inaccurate. Also, the availability of templates doesn't guarantee accuracy. Other methods include Ab Initio Methods and Fold Recognition. Fold recognition comes with the limitations of high computational intensity that skyrockets when processing through a large number of folds. Though many improvements are made in the prediction of protein structures and solvent accessibility, the major challenge in fold recognition is the issue of pattern degeneracy [14,15]. The inherent complexity of protein folding that is influenced by numerous factors like amino acid sequences, environmental conditions, and intermolecular interactions has always made it difficult for computational modelings.

AI (Artificial Intelligence) has truly revolutionized the field of proteomics using DL and NLP. AI has the potential to master intricate patterns (including high-dimensional non-linear patterns) in big datasets, and that is what makes them suitable for a staple computational biology challenge like protein folding. These models use CNN and RNN structures to capture the complex interactions between amino acids [16]. Deep learning models, such as CNNs, hold great promise in proteomics for accurate identification and classification of protein structures, enhancing our ability to detect and understand proteins at a molecular level [63]. AlphaFold: The deep learning model (by DeepMind) relies on abundant sequence data and structure information from the Protein Data Bank (PDB). It integrates CNNs with folding methods to predict amino acid residue distances and dihedral angles and hence establish 3D structures of proteins at great accuracy [17]. This spurred a change in computational biology [18]. By utilizing massive datasets of known protein structures, AlphaFold was able to learn the subtle relationships between amino acid sequences and their corresponding 3D conformations. AlphaFold2, developed by DeepMind, utilizes a sophisticated deep learning architecture that incorporates advanced neural network techniques to predict protein structures. It is capable of understanding the conservation patterns and contextual information of amino acids and has achieved a median backbone accuracy of 0.96 Å r.m.s.d., which is close to experimental accuracy. The model's ability to predict structures of very long proteins (e.g., 2,180-residue proteins) showcases its scalability to a wide range of proteins, addressing a significant challenge in computational structural biology [19]. The accuracy of AlphaFold is illustrated in Fig. 2 [18]. The model's performance in the Critical Assessment of Structure Prediction (CASP) competition, where it outperformed all other methods, highlighted the potential of AI to transform our understanding of protein biology. Building on this effort, the AlphaFold Database Protein Structure Database, first released in 2021 with 300k structures, has accumulated over 214 million predicted protein structures, substantially impacting structural biology [20,21]. Other than AlphaFold, many NLP protein structure prediction techniques include RoseTTAFold, ESMFold, OpenFold, and ProtTrans. With the success of AlphaFold, Baker Lab at the University of Washington developed a framework called RoseTTAFold in the year 2021. This framework incorporated some of the same deep learning techniques but with a different architectural approach. Sidney Lyayuga Lisanza and his team (2024) proposed adapting RoseTTAFold to enable sequence space diffusion. This involves training the model to remove noise while still maintaining the accuracy of the predictions. Their goal was to develop a model that comprehensively understands both sequence and structural feature representations. They represented protein sequences as scale one-hot tensors, introduced Gaussian noise, and progressively refined these sequences and ensured structural prediction accuracy [22].

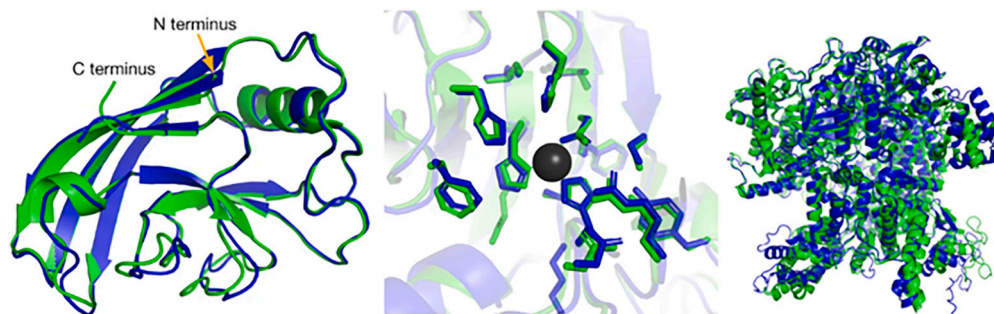


Fig. 2. AlphaFold Accuracy (AlphaFold – Blue, Experiment – Green).

AI's inevitability in the domain of medicine arises from its unmatched ability to adapt and evolve with the rapidly expanding datasets in genomics, proteomics, and other biomedical fields [62,65]. The global healthcare AI market is projected to reach 45.2 billion USD by 2026 [23], and the projected compound annual growth rate (CAGR) for the AI healthcare market is 37%, reaching 188 billion USD by 2030 [24], reflecting the widespread adoption and trust in these technologies. This exponential growth underscores the increasing reliance on AI technologies to streamline operations, enhance patient outcomes, and drive innovations in drug discovery. The volume of medical data is growing exponentially. The National Center for Biotechnology Information (NCBI) reports that the total number of publicly available genomic sequences exceeds 4.3 billion [25]. Traditional analytical methods struggle to keep pace with such data volumes, but AI algorithms, especially deep learning models, can identify intricate patterns and correlations in these datasets. A study published in *Nature* highlighted that deep learning approaches could improve the accuracy of drug discovery predictions by up to 30% compared to conventional methods, significantly reducing the time and cost associated with bringing new drugs to market. Moreover, AI-driven platforms have demonstrated remarkable success in predicting drug interactions and adverse effects, which are critical factors in patient safety. The IBM Watson for Drug Discovery has been utilized to identify potential drug candidates and biomarkers, significantly accelerating research timelines. It has also been used to tailor treatment strategies based on patients' genetic profiles, improving outcomes in diseases like cancer [26]. In clinical trials, AI can streamline patient recruitment by analyzing electronic health records (EHRs) to identify suitable candidates, thereby improving recruitment efficiency by up to 50% [27].

While enormous progress has been made in the field of protein structure prediction, significant challenges remain in the accuracy and interpretability of the predictions. AI-powered models are not transparent and are often referred to as “black boxes” because it is difficult to understand how the model reached its specific prediction. Their lack of explainability restrains their use in high-stake areas such as drug discovery and protein engineering, which require great trust in AI models. To ensure safe and moral developments in drugs, physicians and scientists need to have knowledge on how AI algorithms generate their predictions [28]. Moreover, most currently available methods have an inclination towards primary and tertiary structure prediction, which has caused the secondary structure prediction to be underappreciated, one of the most important and key steps for protein folding and function insights. The pitfalls of the latter demonstrate the significance of models that can do both, i.e., models that not just predict well but also offer transparent interpretable interpretations on how these predictions are made.

Explainable AI addresses the interpretability and explainability of machine learning models, providing transparency and insights into predictive processes [29,30]. XAI methods focus on characteristic analysis and provide a way to identify relevant characteristics according to the intended output. XAI is a technique that aims to explain to the user logically how the corresponding predictions were made. It provides corrective measures, error prediction, and explanations of failures that

occur in the process of prediction. This provides confidence to the researchers [31]. Developing a trustworthy ML model has been one of the most complex challenges to facilitate the use of AI techniques, especially in biotechnology fields. Models that provide interpretability have a significant application in drug discovery [32], examining aspects like thermostability [33], functional classification based on Enzyme Commission numbers [34], and assessing DNA-binding proteins [35].

1.1. Motivation

Our study is motivated by the necessity of developing a more transparent and robust framework for protein secondary structure prediction and improving accuracy. Here we focus on the development and evaluation of an ensemble deep learning model that leverages natural language processing (NLP) metrics and Explainable AI (XAI) techniques. By integrating these advanced approaches, we aim to address key limitations in current methods, such as the lack of interpretability and model transparency, while improving prediction accuracy. The ensemble model combines the strengths of various architectures to create a robust framework for secondary structure prediction. Additionally, the use of XAI methods, including Integrated Gradients and LIME, allows us to gain deeper insights into model decisions and feature importance, further bridging the gap between AI-based predictions and biological interpretability. The results of this study not only showcase the effectiveness of deep learning models in protein structure prediction but also highlight the potential of combining AI with explainability tools to enhance the interpretability and reliability of these predictions. This marks a significant advancement in proteomics, opening doors for further innovations in the integration of AI and bioinformatics.

2. Related works

In recent decades, various computational methods have been developed to predict protein structures. Early approaches relied more on statistical methods. But they produced limited accuracy, especially when applied to different protein datasets. Thus, researchers came up with the idea of neural networks in the field. One of the critical applications of neural networks was conducted by L. Howard Holley and Martin Karplus. In their study, Holley and Karplus developed a neural network model on a dataset of 48 proteins. As shown in Fig. 3 [36], the network model processes an input layer comprising a sliding window of 17 amino acids, encoded upon 21 inputs (representing 20 amino acids and 1 null input). The hidden layer, with 2 units, recognizes the patterns associated with secondary structure (helices or sheets). The model's low predictive accuracy of 63% reflects the model's limited ability to generalize beyond the training dataset of only 48 proteins. Additionally, with only two hidden units, the model's architecture is too simplistic to capture the complex relationships within protein sequences. The binary output layer predicting only helix or sheet structures further limits accuracy by ignoring other significant secondary structures like turns and coils [36].

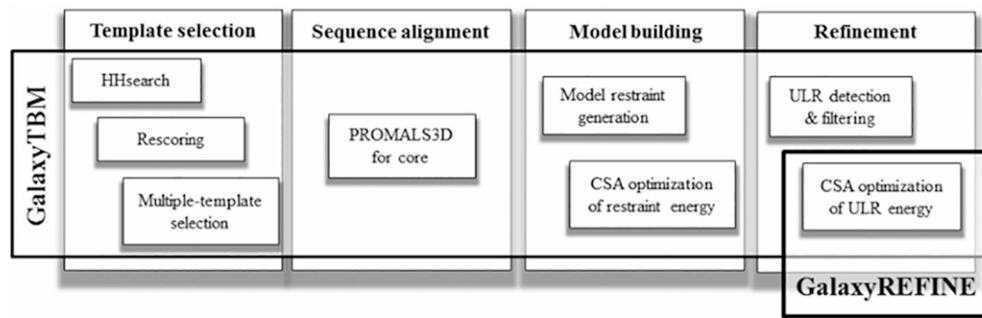


Fig. 3. GalaxyWEB protein structure prediction pipeline.

Petersen et al. made valuable improvements by incorporating evolutionary information through position-specific scoring matrices (PSSMs). Despite the larger dataset of 1,032 protein sequences and the more complex architecture with up to 75 hidden neurons, the model's reliance on fixed sliding windows (15–21 amino acids) constrains its ability to capture long-range dependencies crucial for accurately predicting tertiary structures. Furthermore, the model's accuracy, while an improvement, remains limited to secondary structure prediction in a three-state classification (helix, sheet, coil), which simplifies the complexity of actual protein folding [37]. The GalaxyWEB (developed by Seok Lab) server presents a modern approach with template-based modeling and ab initio refinement methods. This web server refines loop and terminus regions using an optimization-based approach, which was tested in the CASP9 experiment as one of the top-performing methods. The GalaxyWEB server builds upon previous methods by effectively combining multiple templates to generate reliable core protein structures and enhancing accuracy through targeted refinement. The GalaxyWEB structure prediction and refinement procedure is shown in Fig. 4 [38]. However, GalaxyWEB operates as a lighter, more accessible version of Seok Lab's original method, sacrificing some computational rigor for broader usability. This lighter configuration leads to a slight reduction in predictive accuracy compared to Seok's more computationally intensive server, which impacts the results on complex targets due to limited refinement detail and may miss subtle conformational shifts in complex structures.

Marks et al. introduced a method (an integration of computational predictions with NMR spectroscopy and X-ray crystallography) that utilizes evolutionary information included in genomic sequences. This approach involves detecting evolutionary couplings between amino acid residues, which provides critical insights into the functional constraints that guide protein folding and structure. By applying statistical methods to the sequence data, Marks and his colleagues were able to predict the 3-D structures with improved accuracy. This method requires extensive data from multiple sequence alignments and computationally intensive modeling, which can be challenging and resource-intensive and may struggle with disordered regions or proteins with dynamic conformational changes [39]. Yang Zhang proposed and developed a leading tool for automated protein structure prediction and function annotation called the I-TASSER. It employs multiple-threading alignment to identify structural templates from the PDB and constructs full-length models through iterative fragment assembly. The server has performed exceptionally well in assessments conducted by the community at large, placing first in the 7th through 11th CASP contests and outperforming other approaches by 18% with an average AUC score of 0.84 in CAMEO ligand-binding site predictions. It's reliance on template-based modeling reduces accuracy for proteins with limited or no homologs in the Protein Data Bank (PDB) and also restricts its application to novel protein structures, leading to potential errors [40]. András Micsonai proposed BeStSel, a method that focuses on improving protein structure analysis through circular dichroism (CD) spectroscopy. While I-TASSER is a 3D structure prediction tool, BeStSel fills the gap by providing a more accurate protein structure prediction, especially for β -rich proteins. BeStSel

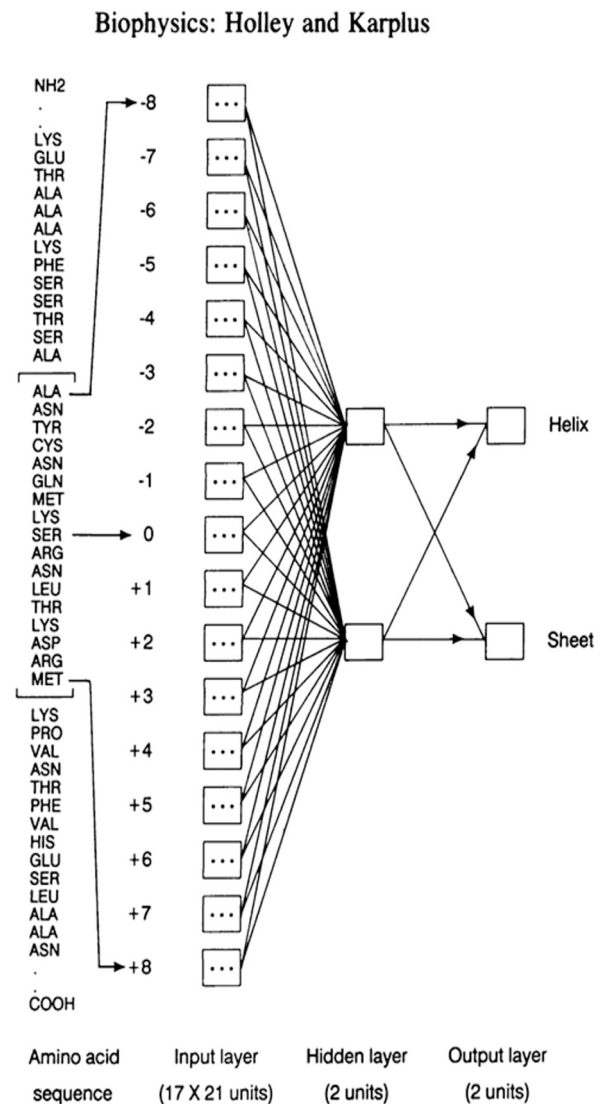


Fig. 4. Neural Network Architecture of L. Howard Holley and Martin Karplus.

estimates by incorporating β -sheet twist and orientation (distinguishing between parallel and antiparallel β -sheets) and evaluating the secondary structure across a broad range of proteins. The method uses a reference set of 73 spectra to optimize eight secondary structure components, including different β -sheet types and helices [41]. But the diversity in β -structure types (especially complex or mixed β -sheet configurations) can lead to ambiguity in structural characterization. Also, BeStSel's prediction accuracy can be impacted by overlapping secondary structure

features in multidomain proteins, and it does miss some nuanced structural variations in the structure.

Further, Zhen Li and Yizhou Yu presented DCRNN, an end-to-end deep neural network that integrates local and global contextual features to predict protein secondary structures. The architecture combines convolutional neural networks (CNNs) with varying kernel sizes for multiscale local feature extraction and a bidirectional gated recurrent unit (GRU) network to capture long-range dependencies. The DCRNN model does multi-task learning to predict secondary structure labels and amino-acid solvent accessibility simultaneously. The model made 69.7% Q8 accuracy on the CB513 benchmark, 76.9% on CASP10, and 73.1% on CASP11. Though it integrates local and global contextual features, capturing very complex, long-range dependencies accurately remains difficult, especially in highly dynamic regions of proteins. The reliance on convolutional layers with set kernel sizes, though effective for local patterns, limits the model's adaptability for long-range context interactions crucial to many protein functions [42]. A major upgrade was brought to the table with the creation of the BeStSel web server by András Micsónai in 2018. This server allows users to upload their CD spectra for quick and reliable secondary structure analysis. The BeStSel algorithm introduced entirely new secondary structure components by accounting for the twist of antiparallel β -sheets. This allows it to distinguish eight secondary structure components. The algorithm performs well in the SRCD range of 175–250 nm but remains effective in the 200–250 nm range used by conventional spectropolarimeters. It also can predict protein folds down to the topology/homology level according to the CATH classification. But for multidomain proteins or those with unusual folds, BeStSel may struggle with precise structure classification, and as it relies on a fixed reference set of spectra, the accuracy is highly dependent on the quality of these reference datasets. Also, its performance in different wavelength ranges varies, where it is less effective outside the preferred SRCD range, limiting its versatility across different experimental setups [43]. Shiyang Long (2019) introduced the Contextnet model to predict the structures by integrating dilated convolutions and concatenation operations. This improves the network's ability to capture both local and global contextual information, leading to more accurate predictions. While the BeStSel was more focused on beta sheets, it wasn't that effective on alpha-helices and other complex structures. However, Contextnet was able to deal with alpha-helices and beta-turns too. When trained on the CullPDB dataset, Contextnet achieved a Q3 accuracy of 84.74% on the CASP13 dataset, outperforming several established models like Jpred and Spider3. The model's effective combination of traditional convolutional neural network features with advanced techniques results in improved prediction performance across various datasets. However, the model's reliance on dilated convolutions, while enhancing local and global context capture, can make it more computationally intensive, potentially limiting its scalability and application in high-throughput scenarios [44].

Richard Evans proposed a method using AlphaFold (developed by DeepMind) using deep learning that directly predicts inter-residue distances rather than just contact points. AlphaFold outperformed all the earlier models with the ability to predict full protein 3D structures with atomic-level accuracy, even for proteins with no prior experimental data or homologous structures in databases. It successfully predicts novel protein folds and disordered regions. This method uses a convolutional neural network (CNN) trained on known protein structure data sets to predict probability distributions of distances between pairs of residues called a distogram. These predictions are used to construct a huge potential of mean force that describes the protein's shape. This is further optimized through gradient descent to generate the final structure. This is further optimized through gradient descent to generate the final structure. Although more precise, this method is computationally intensive due to its optimization process, limiting accessibility for labs without high-performance hardware. It also struggles with highly flexible, multi-domain proteins and unique structures not well represented in existing

databases. This results in potential overfitting and limits the model's applicability to unusual protein topologies [45].

RoseTTAFold (based on the Rosetta framework) was released in the year 2021 by the Baker Lab at the University of Washington, incorporating knowledge from the pre-existing Rosetta framework. RoseTTAFold leverages the same deep learning techniques of AlphaFold, specifically neural networks. Similar to AlphaFold, RoseTTAFold approaches the MSAs (multiple sequence alignments) input feature. Both the frameworks directly learn to predict 3D structures from the amino acid sequences without needing explicit intermediary steps, thus employing end-to-end learning capability [41]. Both AlphaFold and RoseTTAFold are advanced models for predicting protein structures but differ based on specific tasks [46]. AlphaFold is a transformer-based architecture that focuses on attention mechanisms. It was trained on a large dataset derived from the Protein Data Bank (PDB) and uses self-supervised learning techniques to refine its predictions [47]. It uses a model called Evoformer that helps in capturing both evolutionary relationships and spatial interactions between amino acids. It uses a two-step approach that involves a deep learning architecture combining CNN (convolutional neural networks) and attention mechanisms to predict distances and angles between amino acids. AlphaFold offers higher accuracy for individual protein structures. Its streamlined architecture helps in rapid predictions, making it suitable for large-scale studies where speed is a key factor. RoseTTAFold employs a three-track architecture that processes sequences, MSA (Multiple Sequence Alignments) features, and pairwise distances simultaneously. RoseTTAFold uses a combination of both CNNs (convolutional neural networks) and recurrent neural networks, thus helping it to learn both spatial relationships (through CNNs) and sequential dependencies (through RNNs). This mainly helps RoseTTAFold grasp the complex interactions that influence protein folding. RoseTTAFold's architecture allows it to handle a broad variety of protein types and interactions [48]. Though RoseTTAFold is comparatively slower than AlphaFold, it has the strength to handle protein-protein interactions and predict protein structures with a limited amount of evolutionary information available. Yet it struggles with complex, multi-domain, and highly flexible proteins. The accuracy decreases notably for proteins with large, multi-domain configurations or intricate domain interactions. Also, its performance declines when predicting structures for proteins such as de novo or synthetic proteins.

Jinyong Cheng presented a method combining convolutional neural networks (CNN) and long short-term memory (LSTM) models as shown in Fig. 5 [49]. The CNN architecture consists of 2 convolutional layers (max-pooling layer and ReLU activation layer). The feature maps from the second convolutional layer are passed to a softmax classifier to obtain the first probability output. The CNN focuses on extracting local features, and the LSTM captures the sequence context of amino acids, and their combination improves the overall accuracy, achieving a Q3 accuracy of 80.18% on the 25pdb dataset using 3-fold cross-validation [49]. But this method has a limited generalization across different protein datasets. It encounters difficulties managing long-range dependencies in complex sequences, which might reduce predictive accuracy for proteins with intricate secondary structures.

Shusen Zhou introduced a supervised learning method called Combining Deep Neural Networks (CDNN) for protein secondary structure prediction. CDNN is a combination of both convolutional neural networks (CNN) and long short-term memory networks (LSTM). This model uses multiple CNNs with varying layers and filter sizes to extract features from the protein sequences. Then, bidirectional LSTM is employed to continuously extract features based on the raw and CNN-extracted features. Finally, a fully connected dense layer maps these features to different protein secondary structure classes. The model performed best on the CullPDB dataset and CB513 dataset. The model achieved high accuracy specifically for 3 major structure classes (H, E, L), which constitute 73.25% of all samples [50]. However, the model's high computational complexity, varied filter sizes, significant processing power and

Table 1
Comparison of different protein structure prediction models, methodologies, and limitations.

Author and Model Name	Key Features	Limitations
David T. Jones, PSIPRED [53]	Utilizes Neural Networks with Position-Specific Scoring Matrices from PSI-BLAST to predict secondary structures	Produce lower accuracy compared to more recent models like AlphaFold.
Jinbo Xu, RaptorX [54]	Uses deep learning with a contact map that helps in determining spatial relationships between residues.	Complex to train and requires accurate contact map information.
S. Wang, DeepCNF [55]	Uses a Deep Convolutional Neural Network combined with Conditional Neural Fields for prediction	Has a high computational cost and might struggle in the case of long sequences.
D. T. Jones, MetaPSICOV [56]	Combines machine learning and deep learning techniques for contact map prediction and structure prediction.	Has a complex model integration and is slower than single-model techniques.
R. Rao, TAPE [57]	Applies transformer based architectures for predictions, leveraging attention mechanisms	Inaccurate for complex or unusual protein conformations and requires high memory.
K. Kotowski, Spider3 [58]	Uses deep learning with Recurrent Neural Networks (RNN) to predict both the secondary structure and backbone angles of proteins.	Provides lower resolution predictions and heavily relies on the quality and quantity of the input.
A. Rives, ESM-1b [59]	Performs evolutionary sequence modeling using a transformer architecture.	Requires high computational resources for large datasets.
A. Elnaggar, ProtTrans [60]	Uses transfer learning from natural language processing models to model protein sequences, benefiting from pre-trained transformer networks.	Needs fine-tuning for specific prediction tasks and overfits in cases of complex data.
Z. Peng, trRosetta [61]	Uses a deep residual neural network to predict inter-residue distances and orientations, which are then used to generate 3D protein models.	Performance decreases with limited homologous sequences.

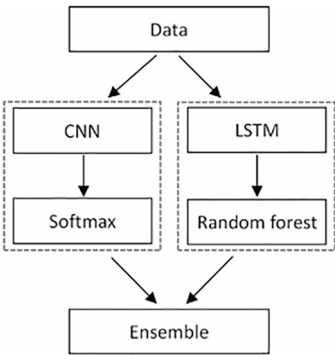


Fig. 5. Prediction Model of the Integrated Deep Learning Model.

memory demands limit its scalability and feasibility for larger datasets or real-time applications.

Vrushali Bongirwar proposed the MCNN-GRNN model, which combines multi-scale convolutional neural networks (MCNN) and gated recurrent neural networks (GRNN) for protein secondary structure prediction. The MCNN component captures local interactions within the protein sequence, while the GRNN component captures long-range dependencies, improving the accuracy of predictions. The robustness of the model is increased by introducing mutations and noise into the subset of protein sequences. This model achieves a decent accuracy in predicting protein secondary structures, with performance metrics evaluated using Q3 and Q8 accuracy scores. Although the model's accuracy slightly decreases when applied to mutated and noisy datasets, it remains competitive with other state-of-the-art algorithms [51]. But, this model remains computational high and its resilience in diverse and noisy biological contexts could be improved for practical applications.

Lufei Zhao introduced a combined model integrating an improved temporal convolutional network (TCN), bidirectional long short-term memory (BiLSTM), and a multi-head attention (MHA) mechanism. This model incorporates knowledge distillation techniques from the ProtT5 pre-trained model. By combining multiscale fusion and bidirectional techniques, the model enhances conventional TCN models and facilitates feature extraction from amino acid sequences. High accuracy rates were obtained when testing performance on multiple datasets (TS115, CB513, and PDB 2018–2020): 88.2% (Q8) and 91.3% (Q3) for TS115, 84.9% (Q8) and 90.3% (Q3) for CB513, and 95.3% (Q8) and 96.8% (Q3) for the PDB dataset. Various models were compared, and the distillation-improved TCN-BiLSTM-MHA model generally outperformed others. For instance, it showed an 8.1% improvement in ACC (Q3) on the CB513 dataset and a 7.9% improvement in ACC (Q8) compared to baseline models. Lufei Zhao and his team also conducted ablation experiments,

which confirmed the added modules' effectiveness, improving model performance as complexity increased. This model, in comparison with CNN-LSTM and Transformer-based models, showed significant improvements across different datasets, particularly in ACC scores. It achieved a maximum improvement of 7.0% (Q3) and 5.9% (Q8) on the CB513 dataset compared to the Transformer model [52]. Additionally, while the model shows accuracy improvements over baselines, the incremental gains may not always justify the added computational cost, especially when simpler models achieve competitive results. Table 1 discusses various other methods that were proposed for the structure prediction of proteins.

It is known that machine learning-based frameworks can assist in discovering emergent patterns and behaviors within complex proteomic data, enabling researchers to predict interactions and functional insights across diverse protein systems [64]. While numerous computational approaches have made significant strides in protein structure prediction, many of them fall short in handling the complexity and variability inherent in biological systems. Traditional models often rely on limited data sources, which can lead to biases and inaccuracies in predictions. Additionally, the lack of interpretability in many methods hinders their practical application in clinical settings, where understanding the underlying decision-making processes is crucial. Furthermore, existing models frequently do not integrate diverse data types, such as evolutionary information and structural features, which can lead to a reduction in predictive power. This highlights the need for more holistic and interpretable approaches that leverage multimodal learning and Explainable AI (XAI) techniques, paving the way for our proposed methodology that aims to bridge these gaps and enhance predictive performance in protein structure prediction.

3. Methodology

In this paper, we employ advanced techniques to predict the secondary structures of proteins using the PS4, CB513, CASP12, and TS115 datasets through a structured and systematic approach. Our methodology consists of three distinct phases that work together to ensure accurate predictions.

The first phase involves thorough preprocessing of amino acid sequences to ensure the data is both reliable and suitable for our predictive models. This process includes normalizing the sequences into a standard format and encoding the amino acids with advanced representations that highlight their biochemical similarities. For example, we use techniques akin to those found in PSSM-based models [37] and beads models for sequence representation, which focus on physicochemical properties. By utilizing encoding methods that reflect these characteristics, we improve the model's capacity to identify subtle relationships within

the data. This strategy has been shown to enhance the learning of intrinsic patterns that were previously missed, as evidenced by similar studies conducted by Rives et al. [39]. In the second phase, we create customized deep learning models specifically aimed at predicting protein secondary structures. These models utilize advanced neural network architectures to effectively recognize and understand the complex relationships between amino acid sequences and their associated secondary structures. We incorporate techniques such as convolutional layers and attention mechanisms, which have shown effectiveness in similar studies like the DCRNN model, where both local and global features are successfully extracted to enhance prediction accuracy [42].

The design of our models closely follows principles found in successful adaptive methodologies, such as SHADE (Success-History-Based Adaptation for Differential Evolution). This approach highlights the significance of maintaining an adaptive framework that can modify itself based on performance feedback. Our model's architecture is subject to iterative evaluation and refinement, mirroring the success history strategies used by Zhang et al. [40]. This resemblance enables the model to learn from its past mistakes and achievements, thereby improving its predictive abilities. The third phase involves rigorous training and optimization of the model utilizing hyperparameter tuning methods to achieve optimal performance levels. Drawing from established practices in deep learning, such as those utilized in the I-TASSER framework, our training process not only adjusts weights and biases but also fine-tunes parameters like learning rates and layer configurations [40]. This carefully managed training regimen ensures peak efficiency in model operation.

Subsequently, we evaluate the trained models using a validation dataset to assess their predictive accuracy and robustness. For validation, we implement metrics alongside the Rouge-L score, similar to the evaluation strategies adopted in the work of Yang Zhang [42]. The Rouge-L metric, which measures the similarity between predicted structures and actual structures, provides nuanced insights into our models' predictive power. This approach not only quantifies performance but also guides further improvements in model design and training processes, reflecting advancements seen in other successful secondary structure prediction methodologies. Overall, this comprehensive approach serves to advance our understanding of protein functionality and dynamics through accurate secondary structure predictions. By integrating state-of-the-art techniques in pre-processing, model development, training, and evaluation, we contribute valuable insights into the biochemical interactions that govern protein behavior. The workflow outlined here, as depicted in Fig. 6, encapsulates the systematic nature of our investigative process, setting the stage for future explorations in protein modeling.

4. Dataset

4.1. PS4 dataset

The dataset comprises a high-quality collection of protein sequences essential for our analysis, providing a robust foundation for predicting protein secondary structures. To ensure model compatibility, the dataset was pre-processed by aligning sequence lengths and numerically encoding amino acids. Each entry includes the starting residue position of the sequence and the amino acid sequence of the protein. Additionally, the secondary structure of each protein is categorized according to the DSSP8 classification, which includes multiple structural types to enhance specificity.

4.2. CB513 dataset

This dataset enables in-depth investigations into protein secondary structures and, like the PS4 dataset, captures unique protein sequences. It includes each protein's amino acid sequence and provides a simplified three-state classification of secondary structure: helix (H), strand (E),

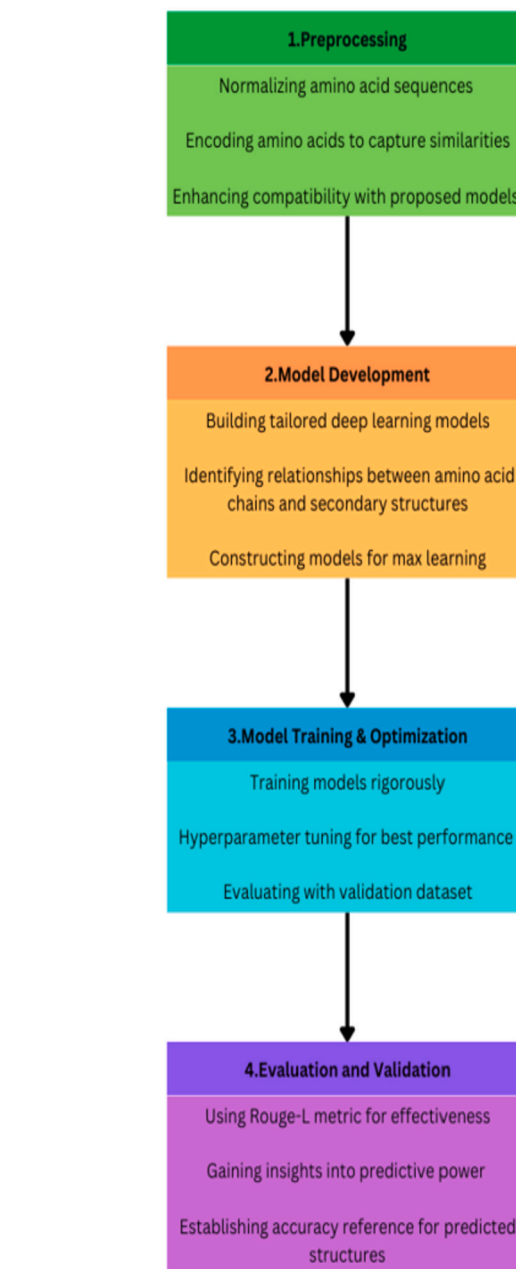


Fig. 6. Workflow for Predicting Protein Secondary Structures Using Deep Learning Models.

and coil (C). Additionally, a more detailed eight-state classification is provided, distinguishing specific types of helices and sheets to offer a comprehensive view of structural characteristics.

4.3. TS115 dataset

This dataset similarly provides structural insights aimed at enhancing the accuracy of secondary structure predictions in models. It includes amino acid sequences and represents each protein's secondary structure using a three-state model (Helix - H, Strand - E, and Coil - C). Additionally, the dataset includes an eight-state classification, offering a more detailed understanding of secondary structure components.

4.4. CASP12 dataset

This dataset is specifically designed for the critical assessment of structure prediction. Organized similarly to the previous dataset, it pro-

vides essential data for prediction tasks, including amino acid sequences for each protein entry, a three-state secondary structure classification (H, E, C), and an eight-state classification. This comprehensive structure enables a more detailed analysis of distinct structural features within proteins. As a whole these datasets provide a diverse foundation for training and the evaluation of the models constructed for the prediction of the secondary structure of the proteins.

5. Data preprocessing

In this paper, we thoroughly examined the dataset from multiple perspectives to determine the appropriate preprocessing methods and ensure that the deep learning models used for prediction are well-suited to the task. Each exploration step plays a crucial role in defining the data characteristics and selecting models for effective training. For each dataset, we calculated the sequence length distribution and assessed amino acid diversity.

Analyzing the sequence length distribution is essential to understanding the dataset's composition. This analysis helps identify the longest chains and ensures that the models can process the entire dataset without truncating sequences, which is critical for preserving the full biological information and avoiding bias. Ideally, the sequence length distribution should form a normal curve, which also reflects effective dataset preparation and sequencing protocols. Any deviations, such as multiple peaks or atypical distributions, may indicate quality issues or contaminants, such as sequence artifacts. In such cases, sequences that are excessively short or long should be trimmed, and quality filtering applied to maintain data integrity and ensure the reliability of downstream analyses.

To further understand the dataset's complexity and anticipate challenges in model building, it is essential to analyze amino acid diversity. Protein sequences consist of various amino acids that define both the structure and function of proteins. Calculating the DIVAA (Diversity in Set of Aligned Amino Acids) scores is valuable for evaluating amino acid diversity, as it measures the spread and frequency of different amino acids within the sequence. This analysis provides insights into biophysical properties and evolutionary conservation, which significantly impact protein secondary structure. Additionally, generative models are often used to simulate amino acid sequences and assess their probabilities within specific protein families. This approach aids in identifying rare variants and hotspot regions that cause mutations—key information for refining predictive models and ensuring they capture the diversity of biological contexts effectively.

The DSSP8 classification, when applied to the secondary structures of protein sequences, provides a nuanced view by detailing specific types of helices and strands. This classification allows for a more granular understanding of protein secondary structures compared to the traditional 3-state classification of helix (H), strand (E), and coil (C). Analyzing the distribution of DSSP8 annotations reveals the structural diversity and complexity within the dataset, providing insights into the prevalence and variety of secondary structure elements. A balanced distribution of DSSP8 states indicates a comprehensive dataset, essential for training models to accurately predict secondary structures across diverse proteins. Conversely, a skewed distribution can signal biases that may impair model performance if left unaddressed. Overrepresentation of certain structures, for instance, can lead to predictive imbalances. To mitigate such biases, we implemented preprocessing strategies such as trimming, normalization, and feature engineering, resulting in a dataset optimized for accurate and reliable secondary structure analysis.

To ensure the integrity and quality of the dataset, the data preprocessing phase is crucial. First, null values are removed to create a clean and robust dataset necessary for subsequent analyses. Next, all protein sequences are converted to uppercase to maintain uniformity. This transformation helps ensure consistent data representation and minimizes potential errors during analysis. Following this, character-level tokenization is applied to both amino acid and DSSP8 sequences

using the TensorFlow Keras Tokenizer. This process breaks down the sequences into individual characters and converts them into numerical representations. It is essential to maintain the correct ordering of the proteins to detect complex relationships within their structures, ensuring that character-level granularity is preserved. In modeling sequences, it is important to maintain their natural order. To achieve this, we employed two padding techniques: post-padding (adding padding at the end of the sequence) and pre-padding (adding padding at the beginning). In this study, post-padding is preferred, as it preserves the natural sequence order, which is vital for models to accurately learn sequential dependencies. This padding was applied to standardize all sequence lengths to the maximum observed length across the datasets, making the dataset compatible with batch processing. By doing so, we avoid truncating valuable biological information. These preprocessing steps lay a solid foundation for developing accurate and reliable models, enhancing performance in predicting the secondary structure of proteins.

The distribution of protein sequence lengths, amino acid frequency in protein sequences, and DSSP8 secondary structure class distribution across the datasets are shown in Fig. 7, Fig. 8, and Fig. 9.

6. Model architecture

Let us understand the mathematical representation of our model. When we include two layers to the LSTM, the output of the forward layer $\vec{b}$ is calculated by using inputs in a positive sequence from $T - N$ to $T - 1$. The output of the backward layer $\overleftarrow{b}$ is calculated by using reversed inputs from time $T - N$ to $T - 1$. We update both the forward $\vec{b}$ and the backward $\overleftarrow{b}$ layers by the following way:

$$b_t = \rho_b (\alpha_{xb} a_t + \beta_{bb} b_{t-1} + q_b) \quad (1)$$

Let us represent the 3 gates (Input, Forget and Output) as I_t , F_t and O_t at time t .

$$F_t = \rho_g (\alpha_f a_t + \beta_f b_{t-1} + q_f) \quad (2)$$

$$I_t = \rho_g (\alpha_i a_t + \beta_i b_{t-1} + q_i) \quad (3)$$

$$O_t = \rho_g (\alpha_o a_t + \beta_o b_{t-1} + q_o) \quad (4)$$

Based on F_t , I_t , and O_t at each time iteration t , the output state $\tilde{c}_t$ is computed as follows:

$$\tilde{c}_t = \tanh(\alpha_c a_t + \beta_c b_{t-1} + q_c) \quad (5)$$

$$C_t = (F_t * C_{t-1}) + (I_t * \tilde{c}_t) \quad (6)$$

Next, the layer b_t , is computed by:

$$b_t = O_t * \tanh(C_t) \quad (7)$$

Thus, the final output of Bi-LSTM is Y_t where,

$$Y_t = [b_{t-n}, b_{t-n-1}, b_{t-n-2}, \dots, b_{t-1}] \quad (8)$$

$$Y_t = \rho_g (\alpha_{by} b_t + C_y) \quad (9)$$

In our study we explored three unique models for the prediction of secondary structure of the protein. Each of the models is explained below.

6.1. Sequence processing powerhouse

The model is designed such that it can capture intricate relations that are embedded within the protein sequences. The sequence processing powerhouse architecture is illustrated in Fig. 10. Different combinations of layers including embedding layer, a bidirectional Long Short Term Memory (LSTM) layer and a dense layer form the building blocks of the complete model. We tailored each component of the model to improve the accuracy of the predictions.

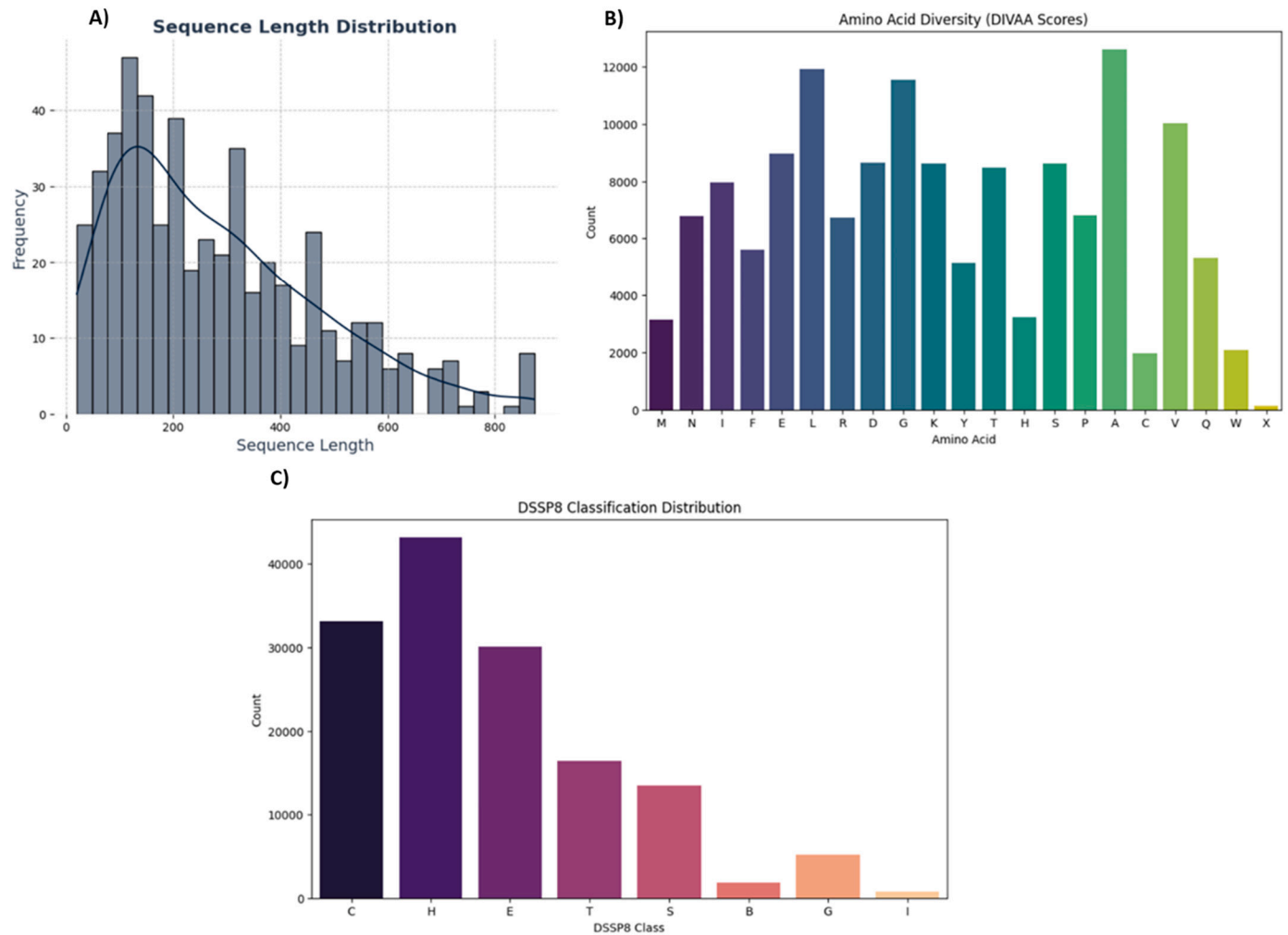


Fig. 7. (CB513 dataset) A) Protein Sequence Length Frequency Distribution B) Amino Acid Frequency (DIVAA Score) C) Distribution of Protein Secondary Structure Classes (DSSP8).

The raw protein sequences are transformed into high-dimensional vectors that encapsulate the biochemical features of the amino acids. This transformation is crucial for exploring the complex interrelations among amino acids and enables efficient handling of extensive datasets common in the bioinformatics domain. In our experiment, we optimized the embedding layer with 1,728 parameters, allowing it to accommodate sequences of up to 1,723 amino acids in length. The primary function of this layer is to enhance the model's ability to analyze a diverse range of protein sequences, underscoring the significance of embedding in this context. In contrast, AlphaFold employs the Evoformer module to capture pairwise evolutionary relationships by processing multiple sequence alignments (MSA) and templates. Similarly, RoseTTAFold utilizes a three-track network that processes sequence data, distance maps, and 3D coordinate information, thereby efficiently learning the relationships between sequence and structure. This differentiation in approaches highlights the varied methodologies in protein structure prediction, emphasizing the importance of tailored techniques for effective analysis.

The next layer in our model is the bidirectional Long Short-Term Memory (LSTM) layer, which is crucial for processing embedded protein sequences. By leveraging both forward and backward sequences, the bidirectional LSTM can effectively capture long-range dependencies within the protein context. This capability is essential for predicting the three-dimensional conformations and functionalities of proteins. In our experiments, we utilized 66,048 parameters to enhance the model's understanding of the sequential con-

text of the sequences. This bi-directional processing allows the model to uncover complex relationships among amino acids, thereby improving its predictive capabilities regarding protein structure and function. In contrast, AlphaFold's structure module employs Graph Neural Networks (GNNs) to represent spatial relationships between residues, implementing attention mechanisms that iteratively refine predictions. Meanwhile, RoseTTAFold utilizes a three-track neural network to simultaneously process sequence data, distance information, and coordinate data, which enables it to grasp intricate sequence-structure relationships. These differing methodologies highlight the diverse approaches in the field of protein structure prediction, emphasizing the importance of tailored architectures for effective analysis.

The final layer in our model is the dense layer, which plays a vital role in translating the learned features into actionable predictions. This layer generates a 10-dimensional vector corresponding to each amino acid in the sequence, representing the probability distribution across various secondary structure classes. The primary purpose of this vector is to enable the model to make robust structural predictions despite the complexity and variability inherent in protein structures. The parameter size of this layer is kept undefined, allowing flexibility in linking high-level feature extraction to final predictions, which is essential within the protein structure modeling framework. In contrast, AlphaFold employs multiple loss functions during the fine-tuning stage, including distance and angle predictions, as well as coordinate predictions as auxiliary tasks. This approach enhances

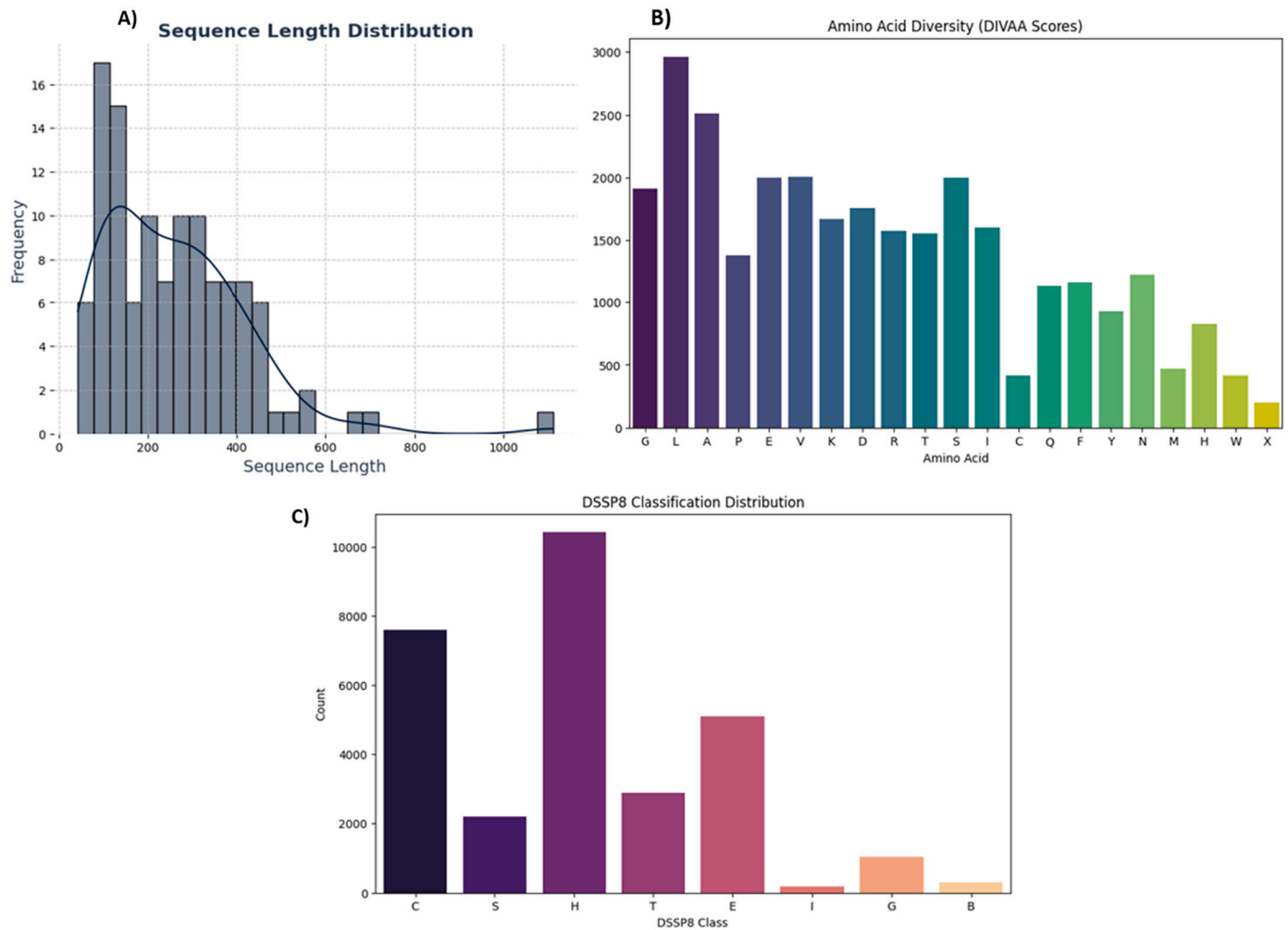


Fig. 8. (TS115 dataset) A) Protein Sequence Length Frequency Distribution B) Amino Acid Frequency (DIVAA Score) C) Distribution of Protein Secondary Structure Classes (DSSP8).

its predictive accuracy. On the other hand, RoseTTAFold leverages all available information from sequences, distances, and coordinates to produce models while being relatively less computationally demanding. These differences in methodology underscore the variety of strategies employed in the field of protein structure prediction, highlighting the importance of optimizing models for both accuracy and computational efficiency.

Overall, when considering sequential complexity, AlphaFold utilizes multiple sequence alignments (MSAs) and template structures to extract critical features from the input information, which are then fed into the Evoformer. In contrast, RoseTTAFold employs a three-dimensional network that incorporates information from sequences, distance maps, and coordinates, facilitating efficient learning of sequence-structure relationships. Our proposed model, on the other hand, captures the complex relationships between amino acids by mapping them into high-dimensional vectors. This approach enables it to effectively learn from large datasets, enhancing its ability to predict protein structures accurately.

6.2. Enhanced sequence-to-sequence predictor

This model is designed to optimize the secondary structure prediction task of the proteins. The enhanced sequence-to-sequence predictor architecture is shown in Fig. 11. The starting layer, that is the embedding layer transforms the amino acid sequences into 64-dimensional vectors to analyze the complex interrelations among the

amino acids that enables the model to navigate and interpret the protein sequences in a high dimensional space. In our experiment, this layer is optimized with 1728 parameters and the embedding layer can handle protein sequences up to 1723 amino acids in length. Thus, this configuration makes the layer compatible to accommodate a wide array of protein sizes and complexities. This type of optimization underscores the capacity of the data to process the extensive data.

The bidirectional layer is essential for processing embedded sequences, as it evaluates data in both forward and backward directions. This layer integrates knowledge and learning from sequential data, enhancing the model's performance. In our experiments, it features 66,048 parameters, which facilitate the contextual understanding of sequences and improve the model's ability to capture long-range dependencies inherent in protein structures. Consequently, this layer effectively captures both local and distant interactions occurring within the protein.

The next layer in our model is the time-distributed layer, which applies a dense operation independently at each position in the sequence. This operation generates a 10-dimensional vector representing the probability distribution across various secondary structure classes for each amino acid. By experimenting with 1,290 parameters, we obtained precise estimates of probabilities for each structural class at every sequence position. This estimation enhances the predictive power of the model and provides detailed insights into the secondary structure of amino acids. The time-distributed layer facilitates a nuanced

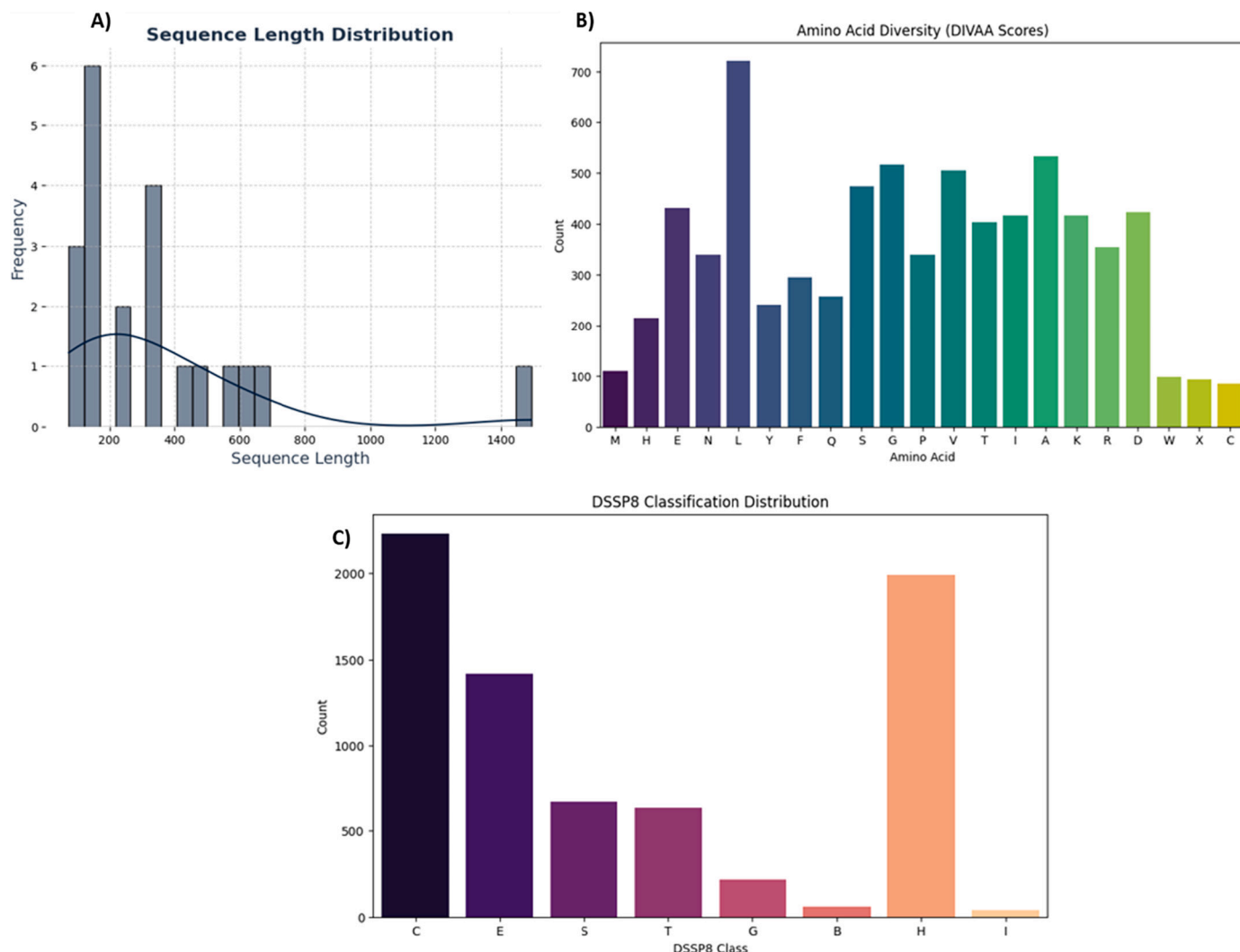


Fig. 9. (CASP12 dataset) A) Protein Sequence Length Frequency Distribution B) Amino Acid Frequency (DIVAA Score) C) Distribution of Protein Secondary Structure Classes (DSSP8).

mapping of the complex nature of proteins. In contrast, AlphaFold fine-tunes its substructures using multiple loss functions and alternative subtasks, such as distance, angle, and coordinate predictions. Meanwhile, RoseTTAFold employs a combination of sequence, distance, and coordinate information to generate accurate probabilistic predictions while utilizing significantly fewer computational resources than AlphaFold.

Overall, AlphaFold employs multiple loss functions and iterative refinement, resulting in highly confident structural predictions that include detailed confidence scores for each prediction. In contrast, RoseTTAFold inductively combines sequence, distance, and coordinate information to generate accurate probabilistic predictions while requiring less computational power than AlphaFold. Furthermore, our proposed model generates a high-resolution probability distribution over all classes of secondary structure at each position, making the time-distributed layer particularly useful for nuanced predictions of structural variability.

6.3. Ensemble model: harmonized sequence predictor powerhouse

In this model, we combined the strengths of both the sequence processing powerhouse model and the enhanced sequence to sequence predictor model. As a result of coupling both the models the overall performance in capturing diverse aspects of the protein

structures is improved. The architecture of this model is shown in Fig. 12.

The input layer serves as the initial stage of the model, preparing raw protein sequence data for further processing. Its primary function is to ensure that all input sequences are formatted consistently, which is crucial for enabling smooth integration and analysis in subsequent modeling steps. Unlike traditional layers, the input layer does not contain parameters; instead, it facilitates a seamless flow of data into the model without altering or processing the input data itself. This role ensures that the input data is well organized and ready for the more complex analysis that follows.

By combining the sequence processing powerhouse with an enhanced sequence-to-sequence predictor, the model can independently process input sequences through their respective architectures. This integration leverages the strengths of both models to better understand protein structures. Each model employs its learning algorithm to create a more holistic interpretation of protein structures, resulting in increased computational speed and improved accuracy.

The next layer in our model is the averaging layer, which plays a crucial role in synthesizing the outputs generated by both the sequence processing powerhouse and the enhanced sequence-to-sequence predictor. This layer averages the predictions from both models, resulting in a unified output that reinforces reliability and coherence in the final pre-

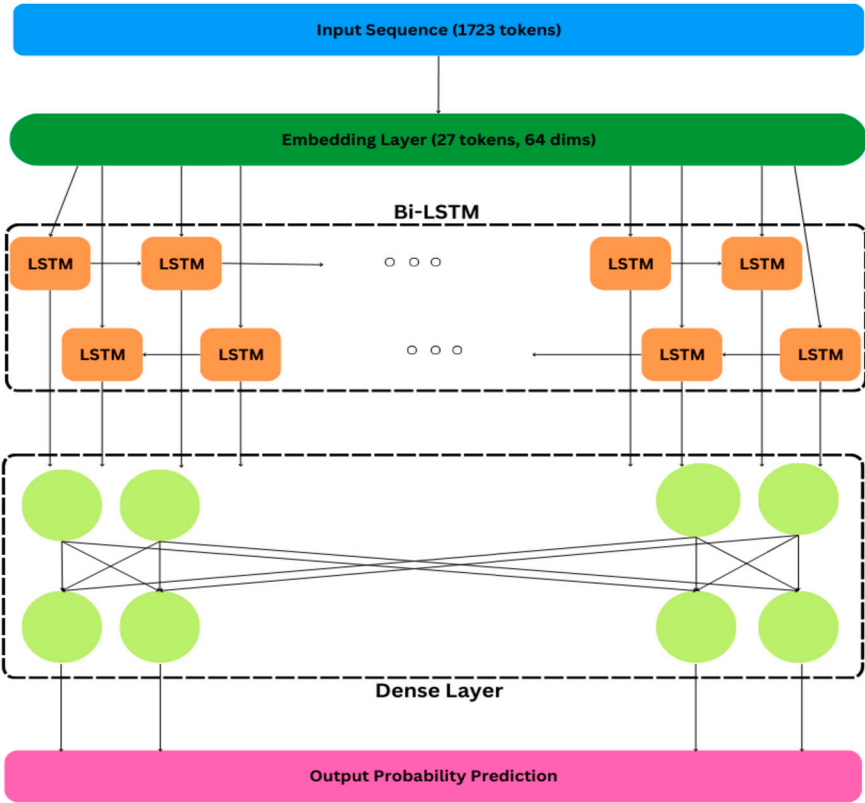


Fig. 10. The Model architecture of Sequence Processing Predictor Powerhouse.

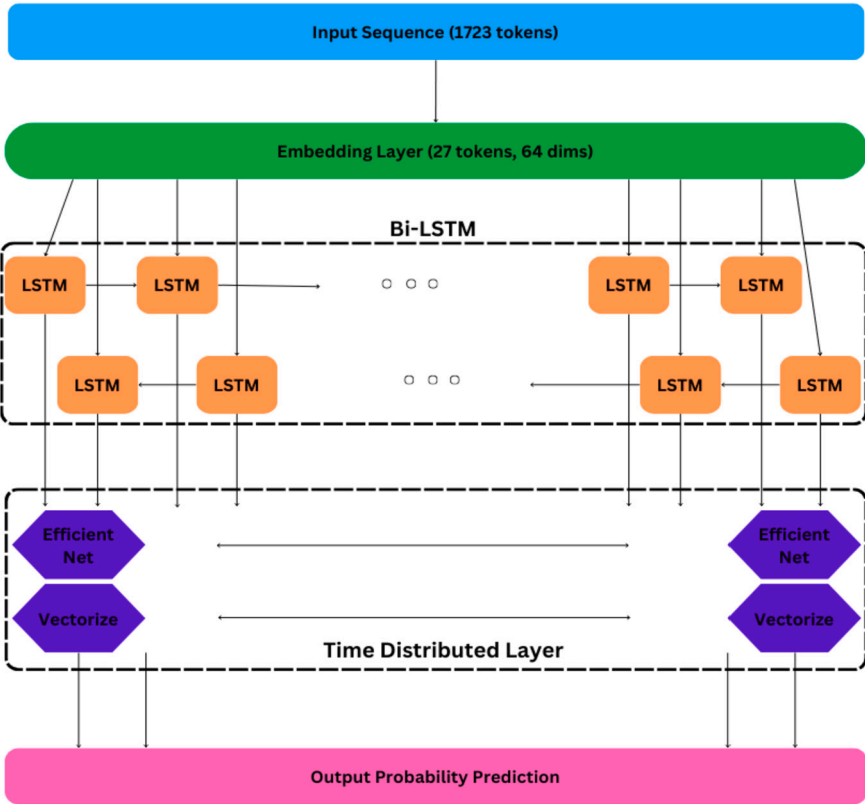


Fig. 11. The model architecture of the Enhanced Sequence-to-Sequence Predictor Powerhouse.

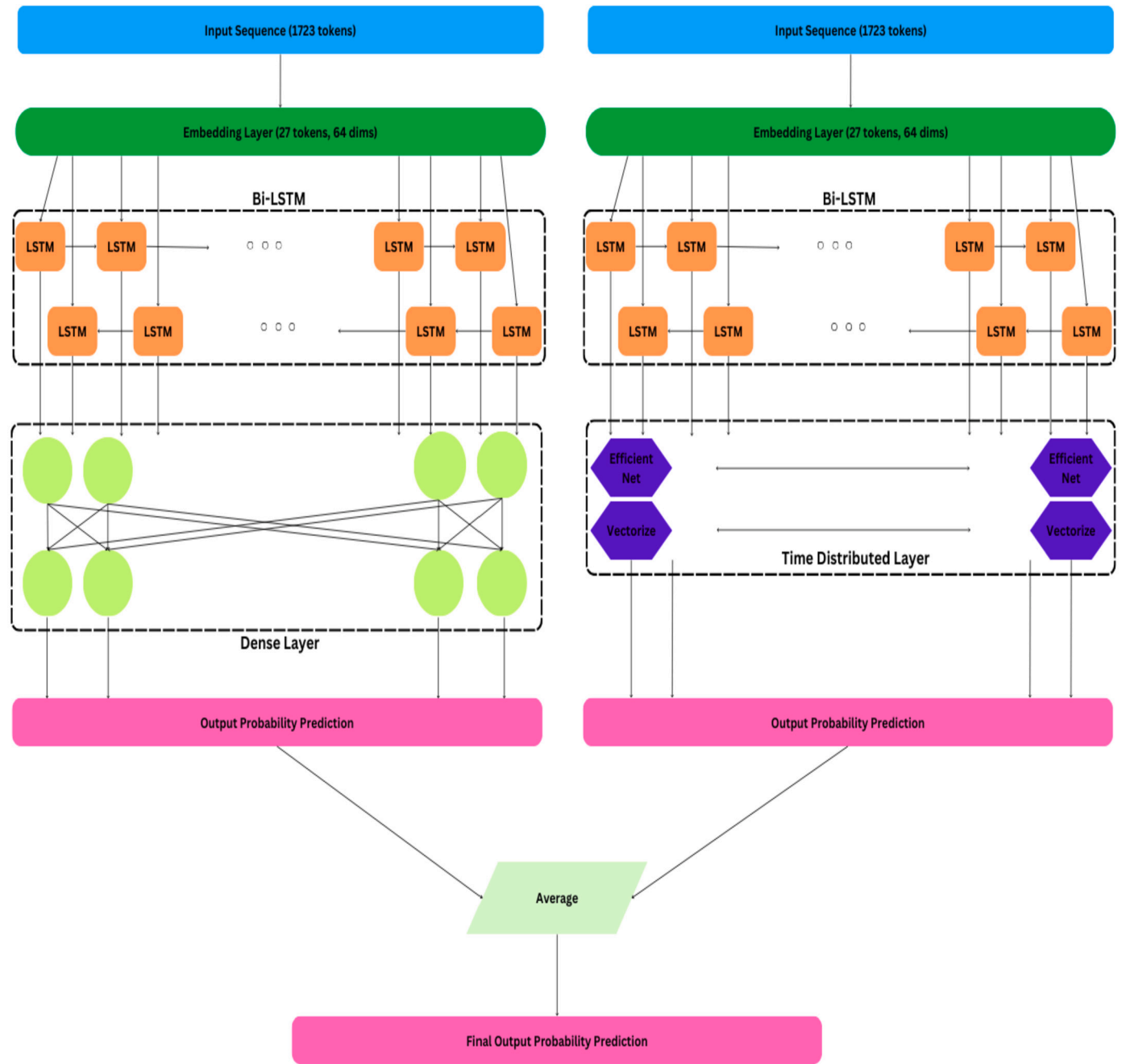


Fig. 12. The Model architecture of Ensemble Model Harmonized Sequence Predictor Powerhouse.

diction. By doing so, the overall bias of the model is reduced, leading to more balanced and dependable results. The ensemble model provides a multifaceted perspective, allowing for a deeper understanding of the sequences. When trained individually, models often risk overfitting, but this issue is mitigated by employing the ensemble technique of averaging predictions. This strategy promotes a more generalized model, as the fusion of both models captures a wide array of sequential and contextual features in protein sequences. Our averaging layer effectively combines diverse insights, balancing the predictions. In contrast, AlphaFold utilizes multiple loss functions during fine-tuning to deliver detailed confidence scores alongside high accuracy. Similarly, RoseTTAFold achieves high accuracy levels through ensemble predictions based on multiple discontinuous crops. This highlights the different approaches taken by various models in enhancing prediction reliability and accuracy.

Once the model is successfully built, we create a pipeline that connects the TF-IDF vectorizer with the models, encapsulating both vectorization and classification steps within a single object. This design ensures that the model can accept raw text as input. Next, we initiate the LIME explainer, which is created by specifying the class name; this component is essential for generating understandable and interpretable explanations of the model's predictions. We select a specific instance from the test set that requires clarification, and the LIME explainer provides insights into the model's probability predictions by perturbing the input instance to create a dataset of similar instances, training a local interpretable model in the process. Our approach enhances the model's interpretability with the LIME explainer, making it more accessible for non-experts to understand its predictions. In contrast, AlphaFold primarily focuses on refining structural and evolutionary information through iterative processes, while

RoseTTAFold employs ensemble and cross-attention mechanisms to predict accurate structures, without an inherent emphasis on local interpretability.

This process begins by selecting a specific data point or instance to be explained, allowing for a focused explanation tailored to that particular instance. The perturbation involves tweaking the features of the selected instance to generate a range of similar but slightly different instances, illustrating how minor changes in the input can significantly impact the model's predictions. In our case, we performed perturbations by randomly altering the words in the sequence of text data. Once the perturbations are made, the complex model is tasked with predicting the outcomes for these altered instances. Each perturbed instance is assigned a weight based on its similarity to the original data point, with instances closer to the original receiving more weight. This weighting reinforces the "local" aspect of Local Interpretable Model-Agnostic Explanations (LIME). Using an exponential kernel, the weighting accounts for the distance between the original and perturbed instances. The kernel width parameter controls how 'local' explanations are determined, affecting the smoothness of the kernel function used for weighting. Smaller kernel width values result in more peaked kernels that focus on points very close to the instance, while larger values create broader, more spread-out weights, allowing for a wider local approximation. This localized model highlights the most important features, making the predictions interpretable. In contrast, AlphaFold utilizes several loss functions and iterative refinement to yield high-confidence structural predictions along with extensive confidence scores. RoseTTAFold combines sequence, distance, and coordinate information to generate accurate probabilistic predictions while requiring lower computational resources than AlphaFold. Our proposed approach, however, returns a probability distribution of secondary structure classes for every position in the amino acid sequence, making it more sensitive to structural variability.

7. Training

For training the ensemble model for the protein secondary prediction, the hardware setup is essential with the involvement of CPUs and GPUs. Our study was performed using Nvidia's GeForce RTX 3080 high-performance GPU that can handle large datasets and complex models efficiently along with ideal parallel processing. Windows OS was utilized on specific requirements. However, Linux can also be preferred for its widespread use in the scientific community. The TensorFlow framework that was developed by Google Brain was used for the building and training process, along with Keras, which is a high-level API that is built on top of TensorFlow. Further, virtual environments are employed for the efficient management of the dependencies by creating isolated environments for each project to prevent the conflicts between the package dependencies.

Algorithm 1 ROUGE-L Metric.

Input: Dataset
Output: ROUGE-L Score
ROUGE-L Score $\rightarrow []$
for input_val, true_val **in** dataset **do**
 (i) model.predict(dataset) $\rightarrow$ predictions
 (ii) argmax(predictions) $\rightarrow$ predictions
 for $i = 1 \dots \text{len}(\text{predictions})$ **do**
 (i) true_seq $\rightarrow$ flatten(true_val)
 (ii) pred_seq $\rightarrow$ flatten(predictions)
 Rouge-L Score $\rightarrow$ fmeasure(true_seq, pred_seq)
 end for
end for

To enhance the training process for our text summarization model, we implemented two custom Keras callbacks: RougeLCallback and

WarmUpScheduler. RougeLCallback, which calculates the ROUGE-L metric on a sampled subset of the validation data, provides a reliable evaluation measure. The WarmUpScheduler addresses the common issue of learning rate instability during early training epochs by gradually increasing the learning rate. These callbacks were integrated into our training pipeline, resulting in improved model performance and faster convergence.

Algorithm 2 ROUGE-L Warmup Evaluator.

Input: Validation Dataset, DSSP8 Tokenizer, sample size = 100
Output: Mean Rouge-L Score
Initialize Validation Dataset, DSSP8 Tokenizer, sample size, Rouge-L Score
On each epoch end:
 a) Input:
 $N \rightarrow$ No. of batches
 $B \rightarrow$ Samples per batch
 b) Unbatching:
 Flatten dataset: $N \times B$
 c) Rebatching:
 No. of batches = Sample Size
 Each batch has 1 sample from the flattened dataset
 d) Compute Rouge-L-i: i^{th} sample Rouge-L Score
 e) Compute Mean:
 $\text{Mean-Rouge-L} = \frac{\sum(\text{Rouge-L}-i)}{N}$
 $N \rightarrow$ total no. of samples

Algorithm 3 Warm Up Scheduler Class.

Input: α, β, η
Output: Updated $\tilde{\alpha}, \tilde{\beta}$ values
Initialize:
 (i) Learning Rate (α)
 (ii) Warmed Learning Rate (β)
 (iii) Warm-up epochs (η)
On each epoch begin:
if epoch $< \eta$ **then**
 $\tilde{\alpha} \leftarrow \alpha + (\beta - \alpha) \times \left(\frac{\text{epoch}}{\eta} \right)$
end if

Where, $\tilde{\alpha} \rightarrow$ updated learning rate

Algorithm 4 Set CallBack Class.

Input: Validation Dataset, DSSP8 Tokenizers, $\alpha = 0.0001, \beta = 0.01, \eta = 5$
Output: Rouge-L CallBack, WarmUp Scheduler, Early Stopping
Call:
 (i) Rouge-L CallBack
 (ii) WarmUp Scheduler
 (iii) Early Stopping
 a) Mode: max
 b) Patience: 3
 c) Verbose: 1
 d) Restore best weight: True

Moreover, the hyperparameter tuning techniques that we utilized played a key role in improving the model performance. Initially, we started with a high learning rate in order to find the maximum value that causes divergence, then reduce iteratively until the model stabilizes. By this way we controlled the step size during each iteration while performing optimization. Generally smaller batches help escape local minima. We used batch sizes of around 10-20 for other specific reasons too. This technique would give better hardware and software optimization. Additionally, the dimension of the embedding layer was fixed to the amino acids that represent the protein sequence. Also, modified the number of layers and units in bidirectional LSTM to improve the memory retention

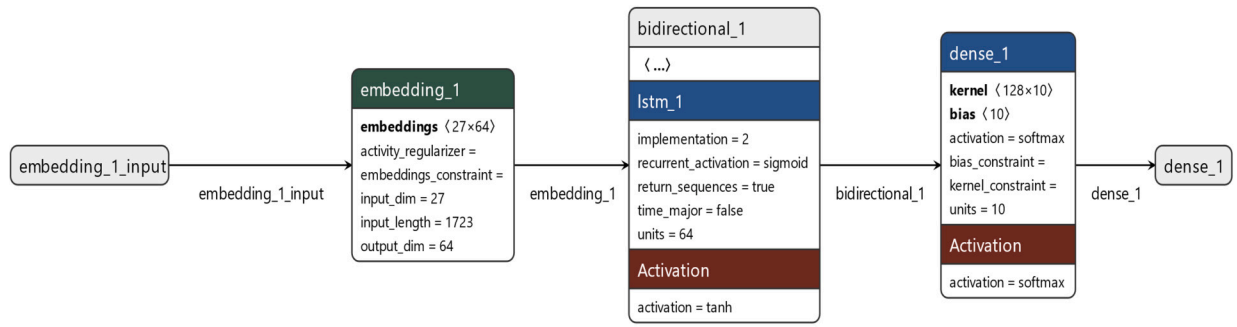


Fig. 13. Layer breakdown and connectivity of Sequence Processing Powerhouse Architecture.

Table 2

Configuration details of Sequence Processing Powerhouse.

LAYER (TYPE)	OUTPUT SHAPE	PARAM #
Embedding	(None, 1723, 64)	1728
Bidirectional LSTM	(None, 1723, 128)	66048
Dense	(None, 1723, 10)	1290
Total params	69066	
Trainable params	69066	
Non-trainable params	0	

Table 3

Configuration details of Enhanced Sequence-to-Sequence Predictor.

LAYER (TYPE)	OUTPUT SHAPE	PARAM #
Embedding	(None, 1723, 64)	1728
Bidirectional LSTM	(None, 1723, 128)	66048
Time distributed	(None, 1723, 10)	1290
Total params	69066	
Trainable params	69066	
Non-trainable params	0	

The loss function uses cross-entropy to measure the mismatch between the predicted values and the actual label. This evaluates the model's confidence in its prediction and quantifies the deviation of the model from the ground truth. This loss function should be minimized to enhance the precision of the model. Whenever the model makes wrong predictions for real-world implications, though the model is highly confident in its output, sometimes lower loss values could be generated for the incorrect predictions.

$$\text{Loss} = - \sum_{c=1}^M y_{o,c} \log(p_{o,c}) \quad (11)$$

To capture the completeness of the predictions of the model via recall and precision ROUGE-L metrics. This metric uses the longest common subsequence method. It measures the similarity between the two sequences by identifying the longest subsequence present in both without requiring contiguity. The ROUGE-L score is calculated using:

$$\text{ROUGE-L} = \frac{\text{Precision} \cdot (1 + \beta_2) \cdot \text{Recall}}{\text{Recall} + \beta_2 \cdot \text{Precision}} \quad (12)$$

Here, the beta (β) represents a weighting factor that adjusts the relative importance of precision and recall. If $\beta = 1$, the formula reduces to the harmonic mean of precision and recall, giving equal weight to both. If $\beta > 1$, recall is given more weight than precision. This means that the model is penalized more heavily for missing relevant information than for generating irrelevant information. If $\beta < 1$, precision is given more weight than recall. This means the model is penalized more heavily for generating irrelevant information than for missing relevant information. A high score indicates that the model performs well in predicting along with maintaining the integral structural representations as represented in Fig. 22, 23, 24.

On integrating all these metrics, a multidimensional evaluation approach is facilitated. By including this metric, the relevance of the model is improved. The model's structural classification capabilities are high using this technique.

This paper aimed to improve protein secondary structure prediction using deep learning models. We evaluated three models on their performance using three metrics: accuracy, loss, and the Rouge-L score. Below, we present the results, followed by a discussion of their significance.

In this view, the great performance of both the Sequence Processing Powerhouse and the Enhanced Sequence-to-Sequence Predictor underscores the very promising results from this research on deep learning models for protein secondary structure prediction. Every model depicted its efficacy of learning, whereby the Sequence Processing Powerhouse model scored a decent accuracy of 93.93% both in training and validation datasets combined at the same time, showing effective learning with minimal overfitting. Validation accuracy and loss values from the Enhanced Sequence-to-Sequence Predictor also displayed high and low values, respectively, indications for high and low confidences and reliability in the predictions. The score for the Rouge-L was high in both models, where at about 0.647, the capacity to make more accurate predictions of structure while maintaining the integrity of the protein se-

and contextual learning. Adapted Adam optimizer for the model optimization tasks. The data was splitted in a ratio of 60% for the training set, 20% for the validation and the remaining 20% for the testing phase. Also the inclusion of different splitting techniques like stratified splitting and timely splitting improved the model performance. In the former splitting, each of the subset is ensured that it maintains in the same class distribution as the entire dataset. Inclusion of this technique avoided biases in case of imbalance datasets. The latter avoids leakage in the data and ensures that the dataset is split into training and validation sets before any preprocessing tasks. Also, the features that were not available in the prediction time were excluded to avoid data leakage and by applying the preprocessing steps individually, the consistency is maintained while training and validation. Then by doing scaling, normalization and encoding steps at each fold during the cross validation, it was ensured that the data does not leak between training and validation datasets. The configuration of the model is mentioned in Tables 2, 3, and 4. The detailed configuration details are illustrated in Fig. 13, 14, 15.

8. Discussion & results

In our study, we evaluated the model's performance with a combination of accuracy, loss and ROUGE-L score (see Figs. 16-24). Let us see each of the models' insights.

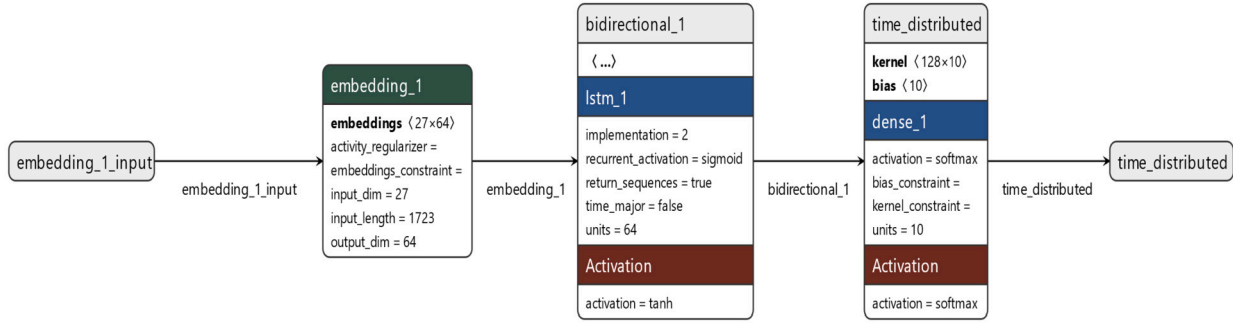
The accuracy metric measures the model's performance in correctly identifying the true classes (alpha-helix and beta-sheet) of amino acids within a sequence. It directly tells the alignment with the realities regarding predictions.

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (10)$$

Table 4

Configuration details of Ensemble Model: Harmonized Sequence Predictor Powerhouse.

LAYER (TYPE)	OUTPUT SHAPE	PARAM #	CONNECTED TO
Input	(None, 1723)	0	-
Sequential	(None, 1723, 10)	69066	Input
Sequential 1	(None, 1723, 10)	69066	Input
Average	(None, 1723, 10)	0	Sequential & Sequential 1
Total params	138132		
Trainable params	138132		
Non-trainable params	0		

**Fig. 14.** Layer breakdown and connectivity of Sequence-to-Sequence Predictor.**Table 5**

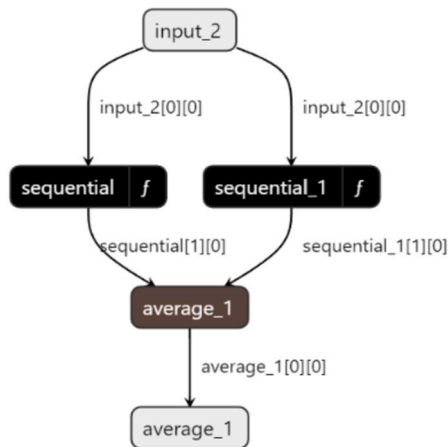
Model Performance across various metrics.

Model Name	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss	Mean Rouge-L Score
Sequence Processing Powerhouse	93.93%	93.93%	0.1708	0.1708	0.6470
Enhanced Sequence-to-Sequence Predictor	93.92%	93.94%	0.1707	0.1704	0.6478
Harmonized Sequence Predictor Powerhouse	94.47%	94.41%	0.1569	0.1585	0.6766

Table 6

Model Evaluation over various datasets (unseen by the model).

Model Name	Accuracy (CB513)	Accuracy (TS115)	Accuracy (CASP12)	Loss (CB513)	Loss (TS115)	Loss (CASP12)
Sequence Processing Powerhouse	88.53%	90.33%	83.50%	0.3117	0.2656	0.4493
Enhanced Sequence-to-Sequence Predictor	88.53%	90.33%	83.55%	0.3115	0.2647	0.4463
Harmonized Sequence Predictor Powerhouse	88.54%	90.39%	83.04%	0.3200	0.2702	0.5276

**Fig. 15.** Layer breakdown and connectivity of Harmonized Sequence Predictor Powerhouse.

quence. The presented ensemble model, however, brought distinguished results, above those of individual performances of the previously discussed, whilst the highest level in validation was equal to 94.41% and in training reached 94.47%. Although this was a small improvement, the

Table 7

Comparison with the existing state of method techniques.

S.NO	MODEL	ACCURACY
1	PSI-Blast based secondary structure PREdiction	75%
2	Jury Of Experts (JOE)	83%
3	Ensemble of PSIPRED, JNet, TRANSSEC	79.1%
4	CNN	85.3%
5	Temporal Convolutional Network (TCN)	84.9%
6	Sequence Processing Powerhouse	88.53%
7	Enhanced Sequence - to - Sequence Predictor	88.53%
8	Harmonized Sequence Predictor Powerhouse	88.54%

benefits indicate good combination from diverse models, supported by smaller loss values between both training and validation phases (0.1569 and 0.1585). Also, its Rouge-L score of 0.6766 reasserts the supremacy of the Ensemble Model, which comes with an outstanding ability of sequencing the order in which the protein structures rely on their biological function as shown in Tables 5, 6. Our results show improved performance compared to the state of art methods as shown in Table 7.

High Rouge-L scores in both accuracy and loss for all models correlate. That presents another indication that would suggest Rouge-L to be a robust metric in the value it has for appraising protein sequence predictions. Its exact focus is on order of sequence and integrity of structure. That is where the biological core of proteins lies: keeping up with struc-

tural integrity. Another study later delves into the interesting parallels of language modeling and biological sequence prediction, highlighted by the successful adoption of the ROUGE-L metric. It lends a new perspective to the view of biological sequences as linguistic structures, whereby the comparison is an ordered alignment of a sequence in the very same way that word alignments in a natural language bring forth deep context and meaning. Moreover, further evaluation over more unseen datasets (CB513, TS115, and CASP12) also gave a promising result for the models in generalization ability and, hence, further strengthened the belief in the improvement of the Rouge-L score through constant experimentation. Since Rouge-L did significantly well over these datasets, it only reinstates the stance for the importance of Rouge-L in bridging the gap between NLP and biological sequence prediction. This work further proved the validation of some models for predicting protein secondary structures and opened new perspectives in bioinformatics research with respect to this attractive interface between natural language processing and the complex human language encoded in proteins of living organisms.

Along with that, to enhance the model performance the LIME was included and obtained results as shown in Fig. 25, 26, 27, 28, 29, 30, 31, 32, 33. The Local Interpretable Model-agnostic Explanations fit a simple interpretable model locally around a specific prediction. This helps the LIME to identify features that are most influential in making that prediction.

LIME can be effectively applied to biological data, including protein structure prediction, enhancing the interpretability and reliability of model predictions in bioinformatics. In our context of protein structure prediction, LIME can be used to interpret the predictions of complex models by identifying which amino acid features or sequence characteristics significantly contribute to the predicted structure. This is especially important for understanding the biological significance of specific predictions. LIME assigns weights to different features in biological models, such as amino acid properties, allowing scientists to understand which characteristics are most influential in determining the outcome of a prediction. Bioinformatics often deals with intricate data and complex models. LIME facilitates the interpretation of these models by breaking down predictions into understandable components, aiding in the detailed analysis of biological data.

Also, LIME can identify significant words or phrases that can make classification decisions and allows for insights that are most impactful. Additionally it also has the ability to detect biases and show how different features influence predictions. When the model disfavors for the individuals based on certain demographic features for the credit scoring. Once the bias is detected, it performs removing, altering of features and feature analysis to uncover the bias and create a potentially robust model. LIME produces feature importance weights that indicate which amino acids or sequence motifs are most influential in the secondary structure prediction. These weights are derived from the coefficients or decision rules of the surrogate model (trained on perturbed instances) and are displayed to show their contribution to the model's prediction. For example, if certain hydrophobic residues consistently result in a higher probability of forming alpha-helix structures upon perturbation, these residues will be assigned higher importance weights. This directly highlights their significance in the model's prediction. The surrogate model provides local explanations for each instance of prediction. By approximating the complex model's decision boundary around a specific instance, the surrogate model (such as a linear regression or decision tree) outlines which features are critical at that local level. This helps in understanding localized behavior of the primary model, thus offering insights into specific predictions.

Using these insights we refined the feature selection by iteratively analyzing the features highlighted as important by LIME. This also reduces the noise and enhances the model performance. Further, this technique identifies and removes redundant features. Overfitting can

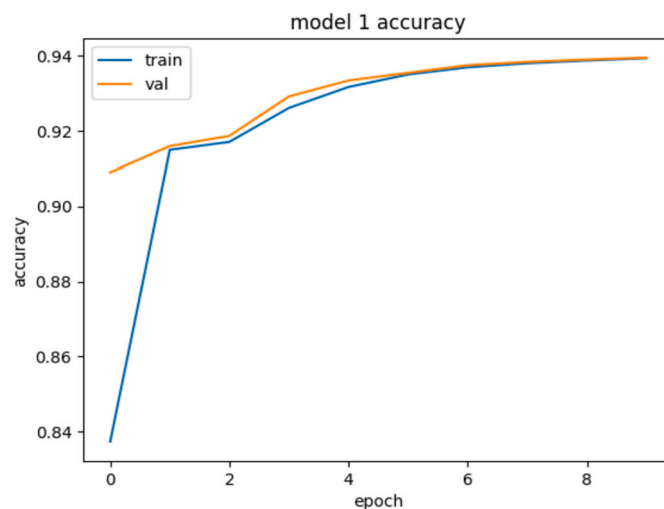


Fig. 16. Sequence Processing Powerhouse train and validation Accuracy curves.

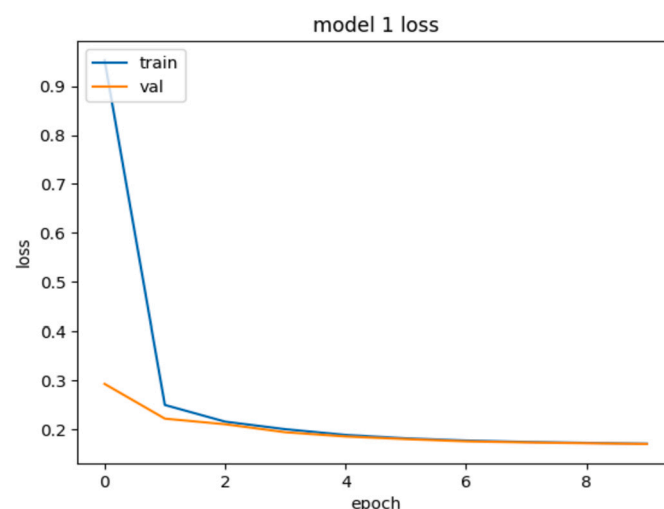


Fig. 17. Sequence Processing Powerhouse train and validation Loss curves.

be reduced by observing which features consistently have low or no impact on the model's predictions. While debugging the algorithms this technique was very helpful as it provides clear and interpretable explanations for each of the individual predictions, thus making a way for retraining.

For a detailed analysis, we visualized the training process over epochs. The graphs clearly demonstrate the models' progression, with accuracy improving and loss decreasing over time, indicating effective learning and generalization as shown in Fig. 16, 17, 18, 19, 20, 21.

Fig. 25 provides valuable insights into the factors driving the Sequence Processing Powerhouse model's predictions for the CASP12 dataset by LIME analysis. The generated feature importance map reveals the key sequence features that contribute significantly to the classification outcomes. The model's most confident prediction is for the class "Other," with a probability of 0.55 and the rest of the classes "s", "b", "c" with 0.12, 0.11, 0.11 probabilities. In this case, one of the "other" probability is underrepresented too. This indicates poor performance of the model for this dataset. To overcome this problem, oversampling was performed to balance the class distribution and to improve the performance of the model.

Fig. 26 provides valuable insights into the factors driving the Sequence Processing Powerhouse model's predictions for the CB513

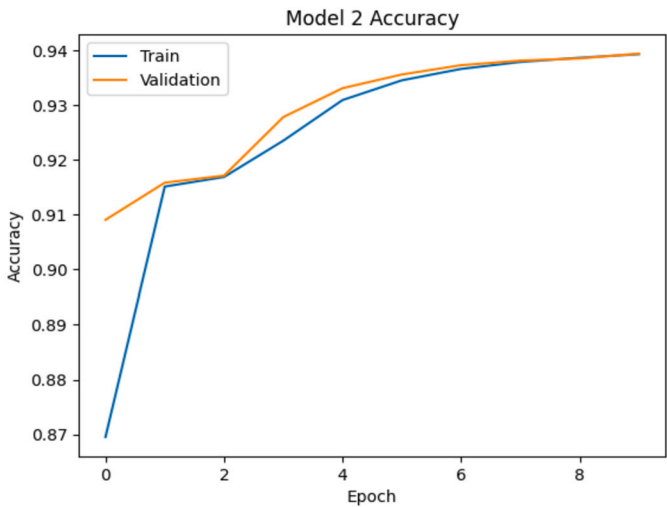


Fig. 18. Enhanced Sequence-to-Sequence Predictor train and validation Accuracy curves.

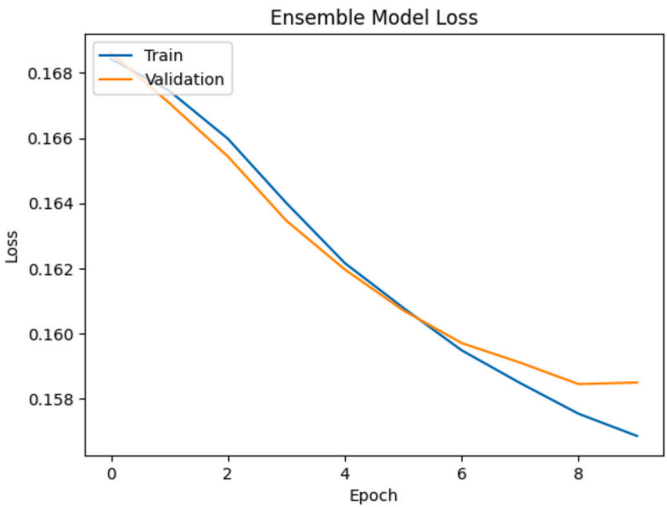


Fig. 21. Harmonized Sequence Predictor Powerhouse train and validation Loss curves.

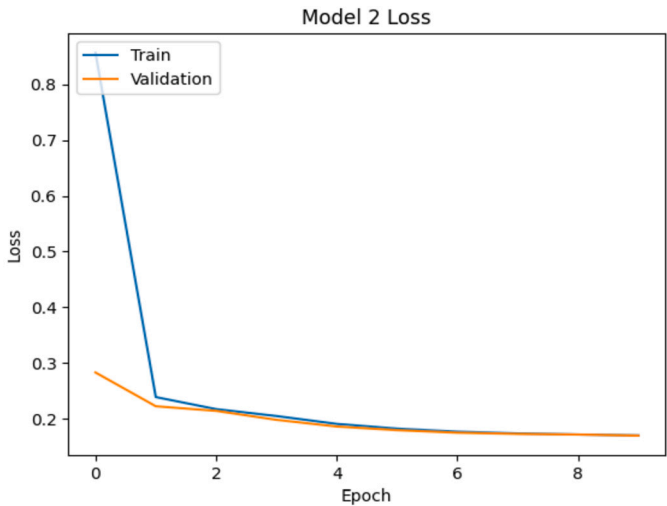


Fig. 19. Enhanced Sequence-to-Sequence Predictor train and validation Loss curves.

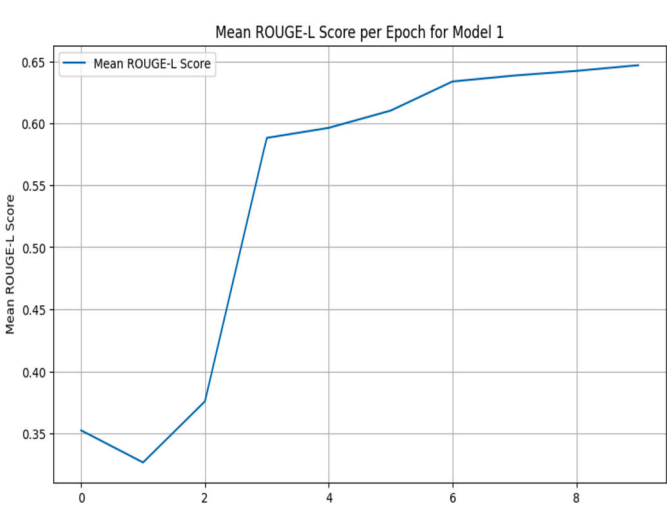


Fig. 22. Sequence Processing Powerhouse rouge-l score.

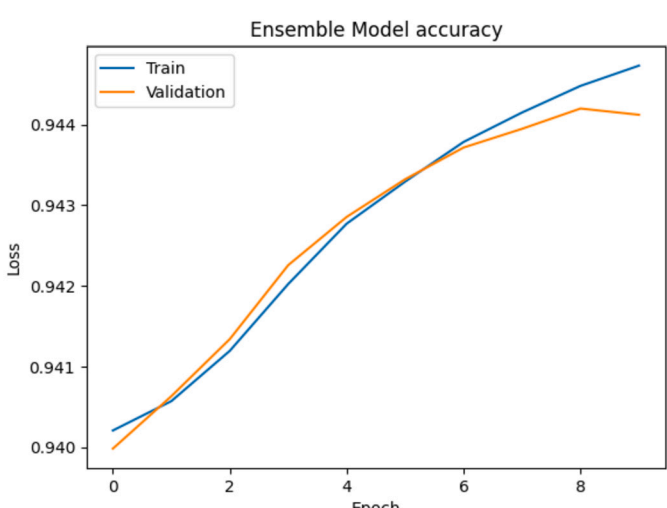


Fig. 20. Harmonized Sequence Predictor Powerhouse train and validation Accuracy curves.

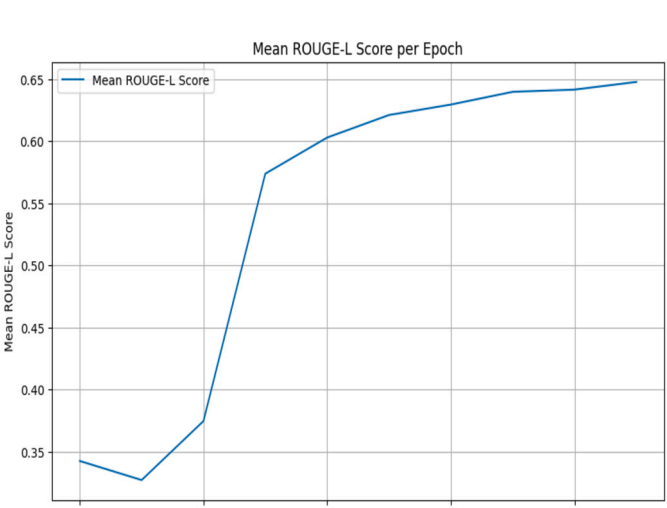


Fig. 23. Enhanced Sequence-to-Sequence Predictor rouge-l score.

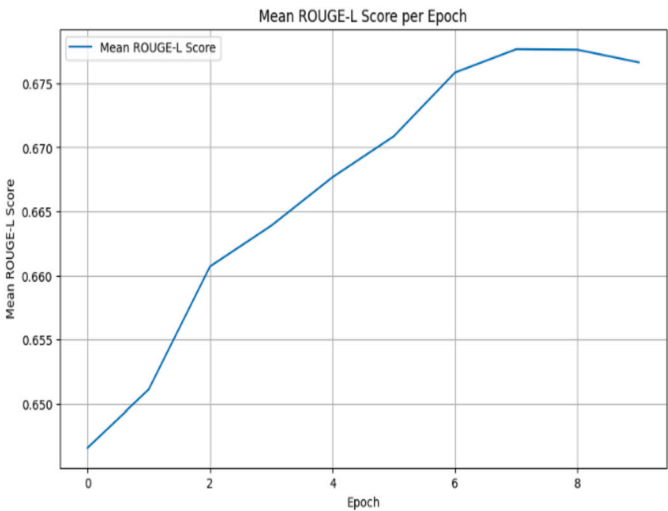


Fig. 24. Harmonized Sequence Predictor Powerhouse Rogue-L Score.

dataset by LIME analysis. The model’s most confident prediction is for the class “Other,” with a probability of 0.59 and the rest of the classes “P”, “b”, “G” with 0.10, 0.10, 0.10 probabilities. In this case, one of the “other” probability is not underrepresented. This indicates that the dataset has uniform class distributions and there is no imbalance. Thus, the model performance is good.

In the same way, Fig. 27 represents that the TS115 dataset trained on SSP has uniform class distribution with classes “b”, “s”, “e” and “c” with probabilities 0.11 each respectively. This is an indicator of balanced dataset and good model performance on that dataset. The highlighted regions within the sequence are identified as the most influential features. The relative lengths of the bars indicate the degree of importance, with longer bars representing more significant features. The LIME analysis underscores the importance of specific amino acids or patterns within the protein sequences in determining the model’s predictions. These findings can be leveraged to gain a deeper understanding of the underlying biological mechanisms and inform future protein structure prediction studies.

We also can observe that the “other” class is underrepresented in Fig. 28, 29, 30, 31, 32 and 33. This suggests that there is an imbalance in the datasets when trained on the Enhanced Sequence to Sequence Predictor and the Harmonized Sequence to Sequence Predictor. Since the

reason for reduced performance is identified with this analysis, we improved it by including data augmentation techniques like upsampling. Further, we performed feature engineering and added structural information like local confrontations including alpha helices, beta-sheets, and random coils that showed a significant improvement in the model performance.

The ensemble approach combines the powers of SPP and ESSP models by leveraging the weaknesses of the individual models. The SPP model exploits the possibility of using evolutionary information and MSA into modeling and so compensates for ESSP model ability to handle representations that do not need MSA. Together, these models make up a robust and rich system for protein structure prediction. Intuitively, integration of these differing models helps to eliminate bias and avoid overfitting through ensembles. Ensemble learning is in general pretty famous for being a more accurate and robust predictor due to its ability to combine more than one model together. This is very essential in bioinformatics, as the complexity of available datasets can be very limited. The ensemble hence achieves enhanced performance metrics with regard to what has been mirrored from the harmonized sequence predictor powerhouse’s performance where, for example, accuracy was improved and also significantly reduced loss values compared to that of individual models.

The ensemble approach is a novel framework as it efficiently connects two sophisticated techniques in protein-structure prediction namely evolutionary data-driven multiple sequence alignment, also referred to as Sequence Processing Powerhouse (SPP); and high-dimensional, sequence-based protein language models, known as Enhanced Sequence-to-Sequence Predictor (ESP). Since each of the models has singularly succeeded in its domain, their synthesis here allows for a holistic approach that brings together the detailed evolutionary insights derived from MSAs with the rapid, large-scale predictive capabilities of protein language models. The novelty also lies in the exploitation of different data sources and representation methods through the approach. While the proposed ensemble combines models relying on different types of information and predictive techniques, it is possible to gain greater accuracy and robustness than single approaches. The overall presentation does make for a more diverse and effective predictor when there are also aspects involving both evolutionary data and sequence-based learning, setting a new benchmark for protein structure prediction. The enhanced performance and generalization capabilities make this model a promising tool in computational biology and bioinformatics, bridging critical gaps between different protein prediction methodologies.

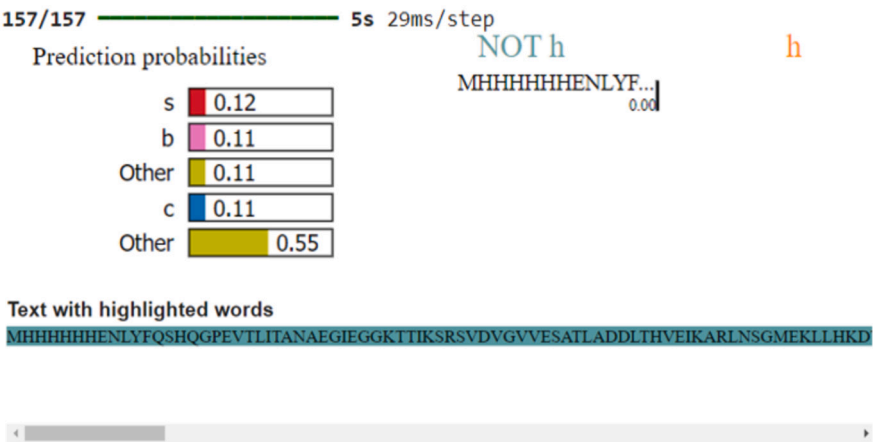


Fig. 25. Feature Importance of CASP12 using Sequence Processing Powerhouse.



Fig. 26. Feature Importance of CB513 using Sequence Processing Powerhouse.

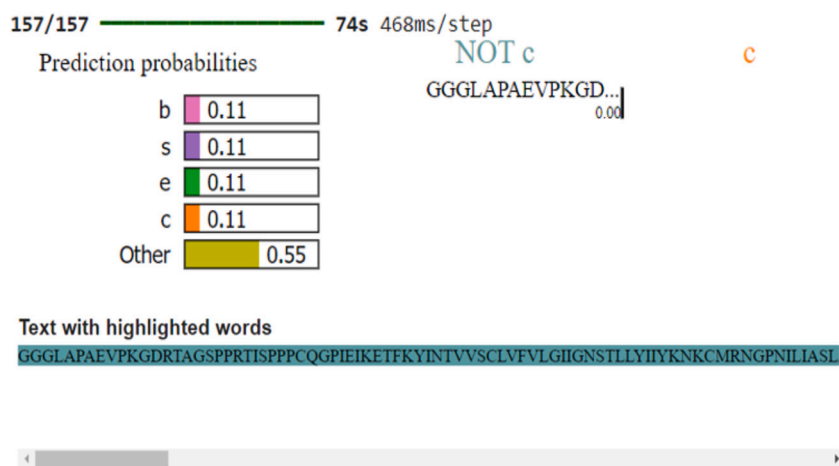


Fig. 27. Feature Importance of TS115 using Sequence Processing Powerhouse.

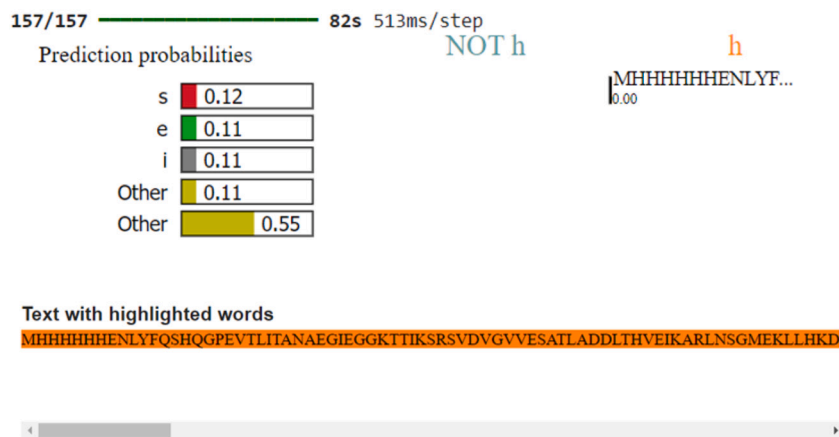


Fig. 28. Feature Importance of CASP12 using Enhanced Sequence-to-Sequence Predictor.

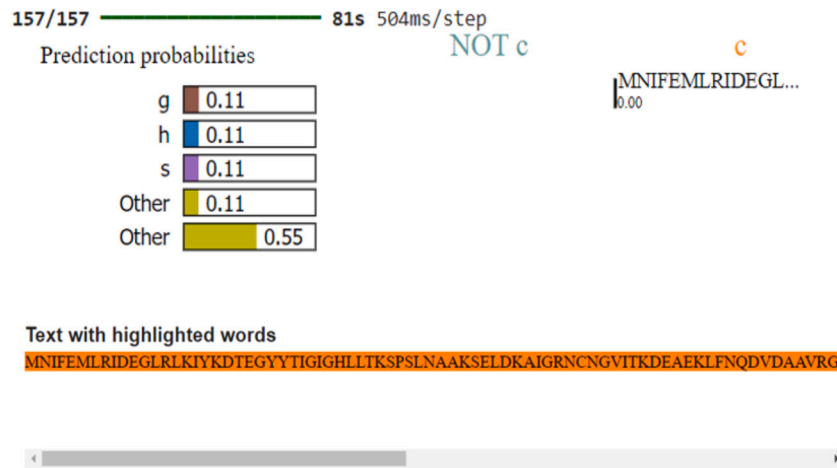


Fig. 29. Feature Importance of CB513 using Enhanced Sequence-to-Sequence Predictor.

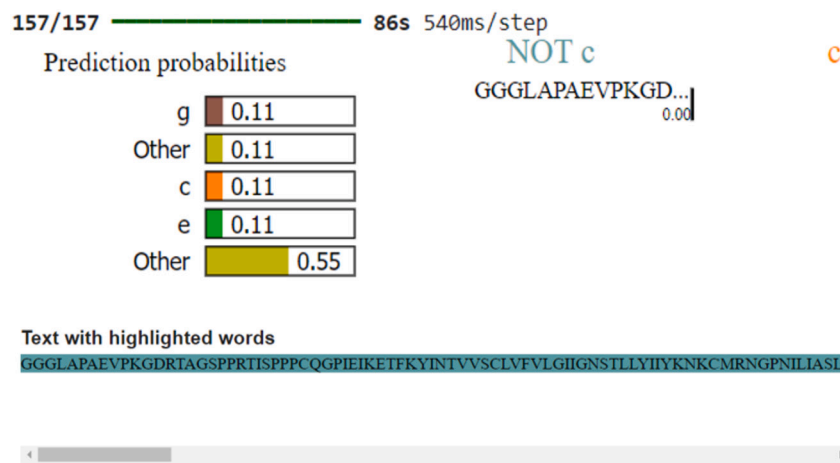


Fig. 30. Feature Importance of TS115 using Enhanced Sequence-to-Sequence Predictor.

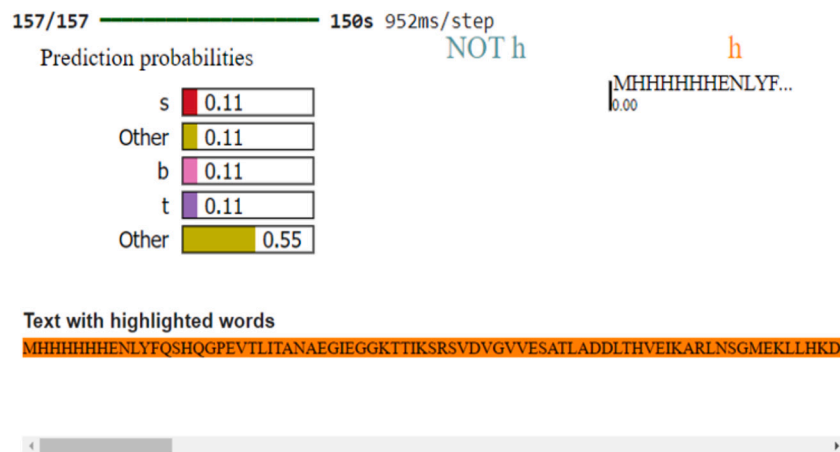


Fig. 31. Feature Importance of CASP12 using Harmonized Sequence Predictor Powerhouse.

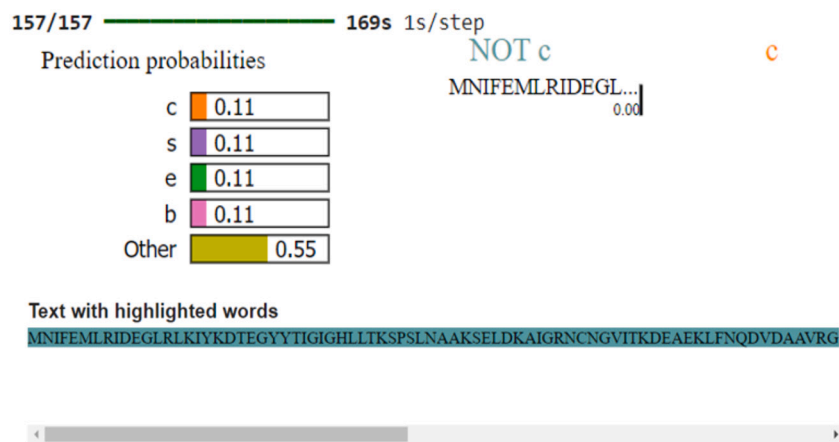


Fig. 32. Feature Importance of CB513 using Harmonized Sequence Predictor Powerhouse.

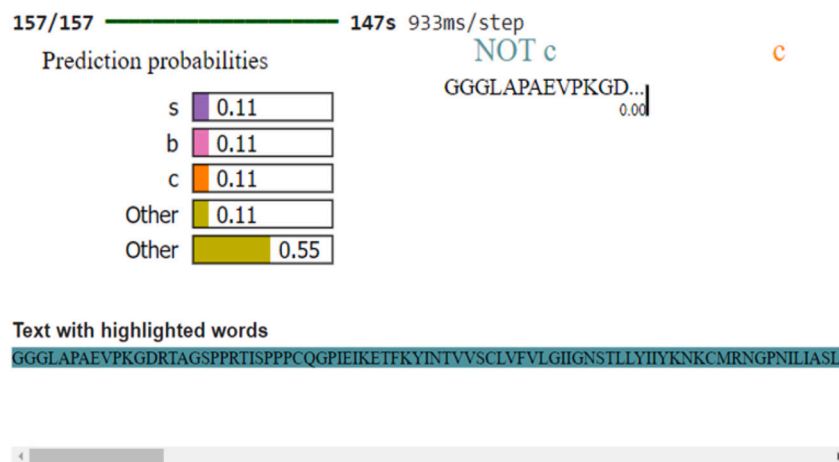


Fig. 33. Feature Importance of TS115 using Harmonized Sequence Predictor Powerhouse.

9. Conclusion

From the research carried out, the deep learning models for protein secondary structure prediction have come out with remarkable results, as each and every model has done well. Among them, the Ensemble Model stands out as the most effective, achieving the highest training accuracy (94.47%) and validation accuracy (94.41%), coupled with the lowest validation loss (0.1585). This consistent pattern of high accuracy, alongside an impressive Rouge-L score, emphasizes the efficacy of these models, particularly in preserving the original protein sequence structures, which is crucial for their biological function. The findings substantiate the validity of the Rouge-L score as a reliable metric for evaluating predicted protein sequences, highlighting its potential in bridging techniques from natural language processing (NLP) into the bioinformatics domain. This integration not only leads to reliable models for predicting protein secondary structures but also opens new avenues for research, facilitating the incorporation of NLP methodologies in bioinformatics. Such advancements deepen our understanding of the “language of life” and its intricate mechanisms.

Furthermore, the application of Explainable AI (XAI) techniques, such as integrated gradients and LIME, marks a significant contribution to enhancing the interpretability of deep learning models in protein structure prediction. By elucidating the critical features that define the relationship between protein sequences and their structures, our study promotes transparency in model predictions and helps

identify potential biases that may influence biological interpretations. This advancement in interpretability is pivotal for understanding the complex interactions within protein sequences and signifies a novel undertaking in the field. Our innovative integration of multimodal learning—leveraging diverse data sources, including evolutionary information, structural data, and protein sequences—distinguishes this work further. This integrative approach enables the capture of richer and more complex features within protein sequences, ultimately leading to more accurate predictions and a better understanding of protein functionality.

The transformative role of AI in medicine and drug discovery is evident and cannot be overstated. AI-driven methodologies enable the analysis of vast datasets, identifying critical patterns and relationships that are often undetectable through traditional methods. By effectively analyzing genomic and proteomic data, AI significantly accelerates drug discovery processes, reducing development timelines and enhancing the likelihood of successful therapeutic outcomes. As AI models predict compound interactions and simulate complex biological processes, they pave the way for personalized medicine tailored to individual genetic profiles, ultimately improving patient care. Additionally, the integration of Explainable AI (XAI) will be crucial in supporting researchers and healthcare professionals moving forward. By providing transparent insights into the decision-making processes of AI models, XAI fosters trust among clinicians and researchers alike. This interpretability enables stakeholders to grasp the underlying factors influencing model predictions, facilitat-

ing the identification of biases and improving the reliability of AI-assisted tools. With XAI, researchers can make more informed decisions based on model reasoning, leading to enhanced patient outcomes.

Given these insights, the collaborative efforts of the 3D consortium are poised to play a pivotal role in advancing our understanding of protein structure and function beyond current capabilities. By harnessing the transformative potential of AI—particularly through our Ensemble Model—we can bridge significant gaps in the drug discovery process and effectively personalize medicine. Our approach enhances predictive accuracy in protein structures while streamlining the identification of drug targets from complex biological datasets. This capability not only reduces development timelines but also improves the likelihood of successful therapeutic outcomes. Moreover, the ethical implications of responsible AI deployment in healthcare can foster more accurate diagnoses, timely interventions, and a comprehensive understanding of intricate relationships within biological systems. As AI continues to evolve, the synergy between human expertise and advanced machine learning techniques will drive innovation in healthcare and drug discovery, ultimately transforming the future of medicine. In conclusion, this research establishes the efficacy of deep learning models in protein secondary structure prediction and emphasizes the broader implications of integrating AI with bioinformatics. Looking ahead, future directions could involve refining model architectures to further improve prediction accuracy, exploring the integration of additional data sources such as metabolomics, and investigating the real-world applications of these models in clinical settings. By advancing our methodologies and understanding, we set the stage for significant breakthroughs in drug discovery and personalized medicine, paving the way for a new era in healthcare that is both innovative and ethically grounded.

10. Symbols and notation

Symbols	Notations
α_{xb}	Weight matrix from input to hidden layer
β_b	Weight matrix between two consecutive hidden states (b_{i-1}, b_i)
q_b	Bias vector of the hidden layer
ρ_b	Activation function to generate hidden state
$\alpha_f, \alpha_i, \alpha_o, \alpha_c$	Weight matrices that map hidden layer input to the 3 gates and input cell state
$\beta_f, \beta_i, \beta_o, \beta_c$	Weight matrices that connect previous cell output state to the 3 gates and input cell state
b_{i-1}	Layer Output
ρ_s	Activation function of the gate
$y_{o,c}$	Encoded label for class c of the observation o
$p_{o,c}$	Predicted probability for class c of observation o
M	Total number of classes in the classification problem
β_2	The parameter controlling the balance between precision and recall

CRediT authorship contribution statement

U. Vignesh: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **R. Parvathi:** Writing – review & editing, Writing – original draft, Supervision, Software, Resources, Methodology, Formal analysis. **K. Gokul Ram:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Kindly refer to the repository link provided

[test_secondary_structure_casp12](#) (Reference data) (GitHub)

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